Plurigenera and canonical algebras in analytic deformations

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11 November 2026

Abstract

We prove invariance of all reflexive plurigenera in proper flat analytic families whose central fiber is compact Kähler and has canonical singularities. The base may be singular or nonreduced, and a single neighborhood suffices for every degree. This proves Kollár's Conjecture 16 with a compact Kähler central fiber and implies Siu's conjecture for smooth families. We also prove invariance of rounded logarithmic plurigenera for smooth families of compact Kähler manifolds with a relative simple normal crossings boundary and real coefficients in $[0, 1)$.

We adapt Levine's infinitesimal extension method to klt central pairs whose log canonical divisor is semiample and whose underlying space admits a compact Kähler modification. An argument with finitely many coherent sheaves gives a common neighborhood and relative semiampleness. The birational reduction uses OpenAI's logarithmic subadditivity, abundance and model theorems. Bingener's deformation theorem and a disk criterion give the passage to arbitrary analytic bases. Over a disk, we prove finite generation, flatness and base change for the relative canonical algebra when the central fiber is canonical and is either compact Kähler or admits a compact Kähler modification and has semiample canonical divisor. Its relative Proj is normal, flat and projective. For smooth disk families whose central fiber belongs to class $\mathcal{C}$ and has semiample canonical bundle, all pluricanonical cohomology dimensions are constant on one neighborhood, simultaneously in every integer tensor degree.

2020 Mathematics Subject Classification. Primary 32G05, 32J27; Secondary 14E30.

Keywords. Plurigenera, deformation invariance, compact Kähler spaces, canonical singularities, relative canonical algebras.

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1 Introduction and main results

For a normal compact complex space with canonical singularities, write

$$\omega_X^{[m]} = (\omega_X^{\otimes m})^{**}, \qquad P_m(X) = h^0(X, \omega_X^{[m]}) \quad (m \geq 1).$$

The deformation invariance of these dimensions is a basic question in complex geometry. Iitaka proved invariance for compact complex surfaces [25]. Siu established it for smooth projective families, first in general type and then in arbitrary Kodaira dimension [47, 48], and formulated the corresponding conjecture for families of compact Kähler manifolds [48, Conjecture 0.4]. Kawamata gave an algebraic proof in general type and proved deformation stability of canonical singularities [26]; Takayama treated projective families with canonical fibers [49]. Păun subsequently simplified Siu's extension argument [44].

For analytic families, Rao–Tsai prove invariance when all fibers are smooth Moishezon manifolds [45, Theorem 1.2(i)] and formulate a conjecture for canonical spaces in Fujiki's class $\mathcal{C}$ [45, Conjecture 1.10]. Kollár proves invariance near a projective canonical central fiber of general type [28, Theorem 1]. We establish the central-fiber theorem below for compact Kähler spaces in every Kodaira dimension. The proof combines a central extension theorem with the logarithmic subadditivity, abundance and birational model results of OpenAI [41, 42].

Theorem A (Canonical central fibers)**.** *Let $f : \mathcal{X} \to S$ be a proper flat morphism of complex analytic spaces, with S not necessarily reduced, and let $0 \in S$. Suppose that X_0 is normal, connected and compact Kähler, with canonical singularities. There is an open neighborhood U of 0 such that all fibers over U have canonical singularities and*

$$P_m(X_s) = P_m(X_0) \qquad (s \in U,\ m \geq 1).$$

If $S = \Delta$, the neighborhood U may be chosen to be a disk such that $\mathcal{X}_U = f^{-1}(U)$ has canonical singularities. In this case, for the restricted family $f_U : \mathcal{X}_U \to U$ and every $m \geq 1$, the sheaf $(f_U)_*\mathcal{O}_{\mathcal{X}_U}(mK_{\mathcal{X}_U/U})$ is locally free and commutes with holomorphic base change, and every section of $\omega_{X_0}^{[m]}$ extends over $\mathcal{X}_U$.

For a projective central fiber, this proves Kollár's Conjecture 16 [28], including Conjecture 17 as a special case. The Kähler hypothesis concerns only the central fiber. The following consequence gives Siu's conjecture for smooth families.

Corollary 1.1 (Siu's conjecture)**.** *Let $f : \mathcal{X} \to S$ be a proper holomorphic submersion of complex manifolds with connected compact Kähler fibers, where S is a connected complex manifold. Then $P_m(X_s)$ is independent of $s \in S$ for every $m \geq 1$. The sheaves $f_*K_{\mathcal{X}/S}^{\otimes m}$ are locally free and commute with holomorphic base change.*

Proof. Theorem A makes each $P_m(X_s)$ locally constant on S, hence constant by connectedness. Grauert's theorem [15] gives local freeness and holomorphic base change. □

For $S = \Delta$, this is [48, Conjecture 0.4]. The twistor family in Section 13 illustrates the fiberwise Kähler hypothesis: its total space is non-Kähler over every nonempty open subset of the base.

1.1 Logarithmic plurigenera

For pairs, Hacon–McKernan–Xu prove invariance of rounded logarithmic plurigenera in projective log-smooth families with coefficients in $[0, 1)$ [18, Theorem 4.2]. Partial results for

ordinary plurigenera in Kähler families include the smooth threefold case [16, Theorem 1.8], extension under nefness hypotheses [9, 6], canonical threefold families with a $\mathbb{Q}$-factorial Kähler total space, and smooth Kähler families of maximal Albanese dimension [11, Theorems 1.4–1.5]. Li–Păun–Wang–Zhou prove the boundary-free result for families with Kähler total space [34, Theorem 1.2]. In the same ambient setting, Liu–Wan prove logarithmic extension and invariance, including real coefficients by degreewise rounding [36, Theorem 1.1 and Corollary 1.2]. Here the fibers may vary in a non-Kähler total space.

Theorem B (Logarithmic plurigenera)**.** *Let $f : \mathcal{X} \to S$ be a proper holomorphic submersion of complex manifolds, with S connected and every fiber compact Kähler. Let*

$$B = \sum_{i=1}^{r} b_i D_i, \qquad b_i \in \mathbb{R}, \quad 0 \le b_i < 1,$$

where $\sum_i D_i$ is a relative simple normal crossings divisor. For $m \ge 1$, set

$$\mathcal{L}_m = \mathcal{O}_{\mathcal{X}}\big(mK_{\mathcal{X}/S} + \lfloor mB \rfloor\big).$$

Then $h^0(X_s, \mathcal{L}_m|_{X_s})$ is independent of $s \in S$. For each $s \in S$, every section of $\mathcal{L}_m|_{X_s}$ extends to a section of $\mathcal{L}_m$ on $f^{-1}(U)$ for some open neighborhood U of s. The sheaf $f_*\mathcal{L}_m$ is locally free and commutes with holomorphic base change. If S is Stein, every fiber section extends over $\mathcal{X}$.

The case $B = 0$ gives another proof of Corollary 1.1. For a fixed m, the rational boundary $m^{-1}\lfloor mB \rfloor$ reduces the logarithmic statement to an integral adjoint in that degree, following [18, proof of Theorem 4.2].

1.2 Extension from a semiample central pair

A *compact Kähler modification* of a normal compact space X is a proper bimeromorphic morphism $\widetilde{X} \to X$ from a compact Kähler manifold. For smooth X, its existence characterizes Fujiki's class $\mathcal{C}$. The extension argument applies in this setting and allows singular klt boundaries.

Theorem C (Extension at a semiample central pair)**.** *Let $g : \mathcal{Y} \to \Delta$ be proper and flat, with normal connected central fiber X, and suppose that $\mathcal{Y}$ is klt near X. Let $\mathcal{B}$ be an effective rational horizontal divisor with coefficients in $[0, 1)$ such that*

$$\mathcal{A} = K_{\mathcal{Y}/\Delta} + \mathcal{B}$$

is $\mathbb{Q}$-Cartier. Suppose that adjunction identifies $\mathcal{A}|_X$ with $A = K_X + B$, where (X, B) is klt, X admits a compact Kähler modification, and A is semiample. For each $m \ge 1$ such that $m\mathcal{B}$ and mB are integral, there is a disk U_m about 0 such that every section of $\mathcal{O}_X(mA)$ extends to a section of $\mathcal{O}_{\mathcal{Y}}(m\mathcal{A})$ over $g^{-1}(U_m)$.

The degree-one construction gives the following companion statement.

Theorem D (Extension of canonical sections)**.** *Let $g : \mathcal{Y} \to \Delta$ be proper and flat, with normal connected central fiber X. Suppose that $\mathcal{Y}$ is klt near X, that X is klt, and that X admits a compact Kähler modification. There is a disk U about 0 such that every section of ω_X extends to a section of $\mathcal{O}_{\mathcal{Y}}(K_{\mathcal{Y}/\Delta})$ over $g^{-1}(U)$.*

Levine developed the infinitesimal extension method for smooth class-$\mathcal{C}$ central fibers carrying a smooth reduced pluricanonical divisor [31, Proposition 1.6 and Corollary 1.8]; see also [32]. The lower-degree argument in his Corollary 1.8 is credited to N. I. Shepherd-Barron. Levine's semiample corollary concerns families whose fibers are all in class $\mathcal{C}$ [31, Corollary 1.10], while his torsion-canonical corollary assumes this only for the central fiber [31, Corollary 1.11]. Kollár–Mori extend the method to terminal singularities [29, Theorem 12.5.5]; Kollár discusses its class-$\mathcal{C}$ scope and Nakayama's projective extension results for canonical singularities in [28, Comments following Conjecture 17]; see [40, VI.3.15–16].

Theorem C extends this method to klt central pairs and their admissible reflexive adjoint powers. Its canonical specialization is Corollary 4.5. The uniformity argument of Lemma 5.1 then gives Corollary 5.2: for a canonical central fiber with a compact Kähler modification and semiample canonical divisor, all plurigenera are constant on one disk and the relative canonical divisor is semiample.

1.3 Canonical algebras and higher cohomology

The same uniformity argument also controls multiplication. It yields a structural form of invariance in both canonical settings.

Theorem E (Relative canonical algebras)**.** *Let $f : \mathcal{X} \to \Delta$ be proper and flat, with normal connected canonical central fiber X_0. Assume that either*

(a) *X_0 is compact Kähler; or*

(b) *X_0 admits a compact Kähler modification and K_{X_0} is semiample.*

There is a disk U about 0 on which the total space and all fibers have canonical singularities. For the restricted family, set

$$\mathcal{F}_m = \mathcal{O}_{\mathcal{X}}(mK_{\mathcal{X}/U}), \qquad \mathcal{R}(f) = \bigoplus_{m\geq 0} f_*\mathcal{F}_m.$$

The disk U can be chosen so that the following hold.

(i) *The $\mathcal{O}_U$-algebra $\mathcal{R}(f)$ is finitely generated and flat, and its graded pieces are locally free. For each holomorphic map $u : T \to U$, there is a natural isomorphism of graded algebras*

$$u^*\mathcal{R}(f) \simeq \bigoplus_{m\geq 0} (f_T)_*(p^*\mathcal{F}_m),$$

where $f_T : \mathcal{X} \times_U T \to T$ and $p : \mathcal{X} \times_U T \to \mathcal{X}$ are the projections.

(ii) *The relative analytic space $\mathcal{C} = \operatorname{Proj}_U \mathcal{R}(f)$ is normal, projective and flat over U. Its formation commutes with holomorphic base change, and*

$$\mathcal{C}_t \simeq \operatorname{Proj} \bigoplus_{m\geq 0} H^0(X_t, \omega_{X_t}^{[m]}).$$

(iii) *If $\kappa(X_0) \geq 0$, there is a proper flat family $g : \mathcal{Y} \to U$ with canonical total space and canonical fibers, and a bimeromorphic correspondence over U inducing a bimeromorphic map $X_t \dashrightarrow Y_t$ for every $t \in U$, such that $K_{\mathcal{Y}/U}$ is relatively semiample. Under* (b)*, one can take $\mathcal{Y} = \mathcal{X}$.*

For a semiample logarithmic central pair, the corresponding adjoint algebra has the same finite-generation and base-change properties (Corollaries 5.3 and 9.1). For a smooth family

whose central fiber belongs to class $\mathcal{C}$ and has semiample canonical bundle, Corollary 9.2 proves constancy of all $h^i(X_t, K_{X_t}^{\otimes m})$, for $i \geq 0$ and $m \in \mathbb{Z}$, on one common disk. The fixed-degree argument uses cyclic covers and Hodge theory; Lemma 5.5 supplies the common disk.

1.4 Proof strategy

The argument over a disk has two parts: extension at a semiample central pair and reduction to that pair by a finite birational program.

For extension, the cyclic-cover construction of Esnault–Viehweg [10, Section 3] identifies a character summand of logarithmic top forms with the required reflexive adjoint sheaf. Deligne's logarithmic degeneration theorem, in the compact Kähler form of [35, Theorem 3.2], gives an injective central period map. Its finite-order versions detect the lifting obstructions, which vanish by Levine's local differential calculation. Proper coherence then produces holomorphic extensions. All the Hodge-theoretic constructions take place on a Kähler modification of the central fiber. For smooth central fibers, a related infinitesimal approach is developed by Cao–Păun, whose extension criteria use central Kählerness and a weighted L^2 condition on the obstruction [5, Theorems 1.1–1.2].

The reduction through minimal models and abundance originates in Nakayama's work [38]. For the disk case of Theorem A, we construct a finite negative program beginning on the prescribed canonical central space. Its steps include the mixed contractions of [28, Definition 7]. We use the analytic cone and contraction theorems of Hacon–Xie [20], relative finite generation [14], and OpenAI's abundance and birational model theorems [42]. The termination proof develops the affine-wall argument in [42, proof of Theorem 3.13]. Contraction extension and Nakayama's theorem deform this finite program. At its semiample endpoint, grouping the canonical powers modulo a generated Cartier multiple gives finitely many coherent sheaves on relative projective space. Their uniform regularity yields one neighborhood for all degrees and controls the canonical algebra.

The passage to an arbitrary analytic base uses deformation openness. Canonical singularities remain canonical [40, VI.5.2], and Bingener's theorem ensures that the nearby fibers remain Kähler [2, 27]. The disk result therefore applies at every point of one fixed neighborhood. A disk criterion, as in [28, paragraph 13], then gives constancy of the entire sequence of plurigenera: coordinate disks establish it on a resolution of the base, and properness descends it to the original base.

For Theorem B, we use the fixed-degree preparation in [18, proof of Theorem 4.2]. Relative crepant stratum blowups make the total and fiber pairs terminal while preserving the chosen boundary denominator; the canonical preparation in [36, Lemma 4.2] is a related construction. We then subtract the fixed boundary in the prescribed degree. This preserves its integrality and excludes boundary components from the contracted loci; it is the fixed-degree counterpart of the divisorial negative-part construction in [36, Lemma 4.3]. The central log program deforms by [20, Theorem 7.7], and Theorem C extends the resulting adjoint section.

Sections 2–5 prove the extension and uniformity results. Sections 6–8 prove Theorem A, and Section 9 proves Theorem E and its cohomological and numerical consequences. Sections 10–12 establish Theorem B. Section 13 gives the twistor examples, and Appendix A records the abundance and nef-comparison results used in the birational arguments.

2 Reflexive adjoints and birational section comparison

2.1 Conventions

We work with complex analytic spaces and holomorphic maps. In statements over a general analytic base, the base and total space may be nonreduced. Holomorphic base changes may also be nonreduced. Elsewhere spaces are reduced, except for the Artinian thickenings used in the extension argument. A normal connected space is irreducible. Families over a disk are considered near the central fiber; shrinking the base includes restricting the total space. We write t for the disk coordinate and use $\mathrm{d}t$ to identify relative and absolute canonical sheaves in meromorphic section comparisons.

For a smooth family, a divisor has *relative simple normal crossings* (relative SNC) if it is locally defined by $z_1 \cdots z_\ell = 0$ in relative holomorphic coordinates.

For a normal space X and an integral Weil divisor D, $\mathcal{O}_X(D)$ denotes the rank-one reflexive divisorial sheaf. We set

$$\omega_X^{[0]} = \mathcal{O}_X, \qquad \omega_X^{[m]} = (\omega_X^{\otimes m})^{**} \quad (m \geq 1).$$

The symbol K_X is additive notation for the canonical reflexive sheaf. Thus, for an integral Weil divisor D and $m \geq 0$, we write

$$\mathcal{O}_X(mK_X + D) := \bigl(\omega_X^{[m]} \otimes \mathcal{O}_X(D)\bigr)^{**}.$$

Relative canonical notation has the same meaning. For a canonical space, the mth plurigenus is $P_m(X) = h^0(X, \omega_X^{[m]})$. A positive degree m is *admissible* for a rational boundary B if mB is integral; the associated adjoint sheaf is $\mathcal{O}_X(m(K_X + B)) = \mathcal{O}_X(mK_X + mB)$.

An adjoint J is $\mathbb{Q}$-Cartier when a positive integral multiple defines an invertible sheaf. Its Cartier multiples determine a holomorphic $\mathbb{Q}$-line bundle. We write $\{J\} = c_1(J)$ for its Bott–Chern class. We write $J_1 \sim_{\mathbb{Q}} J_2$ when sufficiently divisible Cartier multiples define isomorphic line bundles. The adjoint J is semiample if one such multiple is globally generated. For a proper morphism $g : Y \to S$, it is relatively semiample if some Cartier multiple $L = \mathcal{O}_Y(qJ)$ has a surjective evaluation map $g^*g_*L \to L$.

For a holomorphic map $u : T \to S$, write $Y_T = Y \times_S T$, with projections $g_T : Y_T \to T$ and $p : Y_T \to Y$. Base change for a coherent sheaf F refers to the natural map

$$u^*R^ig_*F \longrightarrow R^i(g_T)_*(p^*F),$$

where p^*F is the ordinary sheaf pullback. The space T may be nonreduced. Identifications of these pullbacks with divisorial sheaves on normal fibers will be established by adjunction.

Using compatible local meromorphic canonical frames, we write the discrepancy formula on a log resolution as

$$K_T + B_T = \mu^*(K_X + B) + \sum_Q a(Q, X, B)Q,$$

where B_T is the strict transform and the sum is over exceptional primes. At a nonexceptional boundary prime of coefficient b, set $a = -b$. A pair is canonical, respectively terminal, when all exceptional discrepancies are nonnegative, respectively positive. Klt means that every discrepancy, including the nonexceptional ones, is greater than -1. In particular, a canonical pair with coefficients less than one is klt. Meromorphic computations are made on local Stein neighborhoods; we use descent through proper modifications of normal spaces and finite group quotients.

2.2 Cohen–Macaulayness and restriction

We recall the index-cover proof of Cohen–Macaulayness for divisorial sheaves on klt spaces [30, Corollary 5.25]. Applying it to a normal Cartier fiber gives the restriction isomorphisms needed for the infinitesimal thickenings.

Lemma 2.1. *Let Z be a normal complex space with klt singularities, and let D be an integral $\mathbb{Q}$-Cartier Weil divisor. Then $\mathcal{O}_Z(D)$ is maximal Cohen–Macaulay.*

Proof. The assertion is local. Trivialize a Cartier multiple rD, and form the normalized cyclic index cover. It is finite and étale in codimension one, and its algebra has character pieces

$$\bigoplus_{a=0}^{r-1} \mathcal{O}_Z(-aD).$$

These pieces can be checked in codimension one and then reflexively extended; this is the usual index-cover construction. The discrepancy formula for a finite quasi-étale cover shows that the covering space is klt. It has rational singularities and hence is Cohen–Macaulay. A system of parameters at a point of Z remains a system of parameters at the finitely many points above it. The finite cover algebra is thus a maximal Cohen–Macaulay module downstairs. Taking a character is a direct summand operation in characteristic zero, so every displayed summand is maximal Cohen–Macaulay. Local periodicity modulo rD includes $\mathcal{O}_Z(D)$. □

Lemma 2.2 (Relative restrictions)**.** *Let $g : \mathcal{Y} \to \Delta$ be flat with normal central fiber X, and assume that $\mathcal{Y}$ is normal and klt near X. Let $\mathcal{B}$ be a rational horizontal divisor, let $\mathcal{A} = K_{\mathcal{Y}/\Delta} + \mathcal{B}$ be $\mathbb{Q}$-Cartier, and suppose adjunction gives $\mathcal{A}|_X = A = K_X + B$. If $j\mathcal{B}$ and jB are integral, then*

$$\mathcal{F}_j = \mathcal{O}_{\mathcal{Y}}(j\mathcal{A})$$

is flat over Δ and

$$\mathcal{F}_j/t\mathcal{F}_j \cong \mathcal{O}_X(jA). \tag{2.1}$$

This isomorphism is induced by adjunction at the codimension-one points of X. For $a \geq 0$, put

$$R_a = \mathbb{C}[t]/(t^{a+1}), \qquad X_a = \mathcal{Y} \times_\Delta \operatorname{Spec} R_a, \qquad F_{j,a} = \mathcal{F}_j/t^{a+1}\mathcal{F}_j.$$

These are flat R_a-sheaves, and for $a \geq 1$ there is an exact sequence

$$0 \longrightarrow F_{j,a-1} \xrightarrow{t} F_{j,a} \longrightarrow F_{j,0} \longrightarrow 0. \tag{2.2}$$

For $\mathcal{B} = 0$ and $j = 1$, these identifications give $\mathcal{O}_{\mathcal{Y}}(K_{\mathcal{Y}/\Delta})|_X \simeq \omega_X$.

Proof. A local meromorphic canonical frame represents $j\mathcal{A}$ by an integral $\mathbb{Q}$-Cartier Weil divisor. Lemma 2.1 therefore makes $\mathcal{F}_j$ maximal Cohen–Macaulay. The parameter t is a nonzerodivisor: the sheaf is torsion-free on the normal total space and every component under consideration dominates the disk. Torsion-freeness over the local discrete valuation ring of the base implies flatness.

The quotient by t is Cohen–Macaulay of full central dimension and generically has rank one. It is therefore torsion-free and S_2, hence reflexive on X. At a general codimension-one point of X, the fiber is smooth. Flatness and smoothness of that fiber imply that g is smooth near the point. The relevant integral divisors are Cartier there, and the usual adjunction and boundary restriction identify the quotient with $\mathcal{O}_X(jA)$. Since both sheaves are reflexive, this proves (2.1) globally. This argument also proves the canonical-sheaf assertion directly.

Flatness supplies the displayed Artinian quotients and (2.2). Equivalently, multiplication by t embeds $\mathcal{F}_j/t^a\mathcal{F}_j$ into $\mathcal{F}_j/t^{a+1}\mathcal{F}_j$ with quotient $\mathcal{F}_j/t\mathcal{F}_j$. All identifications are compatible with restriction and multiplication by the parameter. □

Remark 2.3. Equation (2.1) identifies the quotient itself with the reflexive adjoint. This is the restriction used in the Artinian exact sequences (2.2).

2.3 Birational comparison in every admissible degree

Lemma 2.4. *Let (Z_1, C_1) and (Z_2, C_2) be normal canonical pairs with effective boundaries, related by a proper bimeromorphic correspondence. Assume that their boundaries agree on the common nonexceptional prime divisors. Let $m \geq 1$ with mC_1 and mC_2 integral. On a common log resolution $p_i : W \to Z_i$, let C_W be the union of the strict transforms of the two boundaries, with their prescribed coefficients. Then, in the meromorphic pluricanonical identification,*

$$(p_i)_*\mathcal{O}_W\big(m(K_W + C_W)\big) = \mathcal{O}_{Z_i}\big(m(K_{Z_i} + C_i)\big), \qquad i = 1, 2. \tag{2.3}$$

The statement holds on analytic total spaces over a disk, with relative canonical divisors in place of the absolute ones.

Proof. For each i, the discrepancy formula is

$$K_W + C_W = p_i^*(K_{Z_i} + C_i) + E_i, \qquad E_i \geq 0, \quad E_i \text{ is } p_i\text{-exceptional.} \tag{2.4}$$

Canonicity supplies the nonnegative exceptional discrepancies. A boundary component present only on the other model adds a nonnegative coefficient to the corresponding exceptional error. The nonexceptional coefficients agree by hypothesis.

A local section downstairs has an effective $\mathbb{Q}$-Cartier zero divisor relative to the integral adjoint $m(K_{Z_i} + C_i)$. Its pullback has effective zero divisor after addition of mE_i. Hence the pulled-back meromorphic m-canonical form lies in $\mathcal{O}_W(m(K_W + C_W))$. The resulting divisor is integral because W is smooth and mC_W is integral.

Conversely, a section upstairs descends meromorphically and satisfies the required pole bounds at every nonexceptional codimension-one point of Z_i. The rank-one reflexive sheaf downstairs is determined by these bounds, proving the reverse inclusion. Over a disk, tensoring with a local generator of the base canonical line identifies the relative calculation with the same absolute calculation. □

Over a disk, the total-space identity transports an extended section along the correspondence. Its central value is determined on the common dense smooth open set and hence everywhere by reflexivity.

2.4 Persistence of an isomorphism locus

The following elementary application of Nakayama's lemma will be used in Sections 8 and 11 to propagate central isomorphism loci through deformed birational diagrams.

Lemma 2.5. *Let $h : \mathcal{Y} \to \mathcal{Z}$ be a proper morphism over Δ, with $\mathcal{Y}$ flat over Δ. If the induced morphism $Y_0 \to Z_0$ is an isomorphism over an open subset $U_0 \subset Z_0$, then h is an isomorphism over a neighborhood of U_0 in $\mathcal{Z}$.*

Proof. Properness and upper semicontinuity of fiber dimension make h finite near each point of U_0. Write $B \to A$ for its finite local algebra there. The central hypothesis gives $B/tB \simeq A/tA$, so Nakayama's lemma makes $B \to A$ surjective. Let I be its kernel. Since A is flat over the base discrete valuation ring, reduction of $0 \to I \to B \to A \to 0$ modulo t is left exact. Thus $I/tI = 0$, and Nakayama's lemma gives $I = 0$. The resulting neighborhoods cover U_0. □

3 Central period injections on Kähler modifications

The extension argument uses an injective period map on the central fiber. We construct it on a compact Kähler modification and identify it intrinsically on a smooth open subset. The Hodge-theoretic inputs are logarithmic de Rham comparison [8, Proposition 3.1.8] and E_1-degeneration for compact Kähler manifolds [35, Theorem 3.2]. For a compact Kähler n-fold Z and an SNC divisor E, they give the natural injection

$$H^1(Z, \omega_Z(E)) \hookrightarrow H^{n+1}(Z \setminus E, \mathbb{C}). \tag{3.1}$$

Naturality of this map under modifications will give the central injections needed in Section 4.

3.1 The canonical degree

Lemma 3.1. *Let X be a normal connected compact complex space with klt singularities. Suppose there is a proper bimeromorphic morphism from a compact Kähler manifold to X. Put $U = X_{\mathrm{reg}}$, $n = \dim X$, and $\omega_X = \mathcal{O}_X(K_X)$. Restriction, inclusion of the top holomorphic de Rham term, and de Rham comparison define an injection*

$$\mathrm{per}_{1,0} : H^1(X, \omega_X) \longrightarrow H^1(U, \omega_U) \longrightarrow H^{n+1}(U, \mathbb{C}). \tag{3.2}$$

Proof. Principalize the pullback ideal of $\operatorname{Sing} X$ on the given smooth Kähler modification by projective blowups. This gives a proper modification $p : Z \to X$, with Z compact Kähler, such that

$$E = \left(p^{-1}(\operatorname{Sing} X)\right)_{\mathrm{red}}$$

is an SNC divisor. If X is smooth, take $E = \varnothing$. Put $W = Z \setminus E$. The restriction $h : W \to U$ is a proper modification between smooth spaces.

The sheaf calculation is

$$p_* \omega_Z(E) = \omega_X. \tag{3.3}$$

For a local section u of ω_X, let D_u be its effective $\mathbb{Q}$-Cartier zero divisor. At a prime divisor Q on Z, pullback gives

$$\operatorname{ord}_Q(p^* u) = v_Q(D_u) + a(Q, X, 0).$$

The order is nonnegative at nonexceptional divisors. At an exceptional divisor it is an integer greater than -1, by the klt inequality, and is again nonnegative. Thus $p^* u$ is holomorphic.

Conversely, let σ be a section of $\omega_Z(E)$ over $p^{-1}(O)$, where $O \subset X$ is open. Since h is a proper modification of smooth spaces, Hartogs extension gives $h_* \omega_W = \omega_U$. Hence σ descends on $O \cap U$, and $\omega_X = j_* \omega_U$, for $j : U \hookrightarrow X$, by reflexivity. This proves (3.3); the identification is pullback of canonical forms.

The Leray edge map and (3.1) give the injection

$$H^1(X, \omega_X) \hookrightarrow H^1(Z, \Omega^n_Z(\log E)) \hookrightarrow H^{n+1}(W, \mathbb{C}). \tag{3.4}$$

The sheaf identity is ordinary pullback over U. Naturality of restriction and de Rham comparison identifies (3.4) with $h^* \circ \mathrm{per}_{1,0}$. Hence $\mathrm{per}_{1,0}$ is injective. □

3.2 Selected characters of a cyclic cover

For higher adjoint powers, the cyclic-cover construction of Esnault–Viehweg [10, 3.5, Corollary 3.11 and Lemma 3.16] replaces ordinary canonical forms by a character eigensheaf. We first construct the cover on a compact Kähler model, then identify this eigensheaf by a divisorial calculation.

Lemma 3.2. *Let X be a normal connected compact complex space admitting a proper bimeromorphic morphism from a compact Kähler manifold. Let (X, B) be klt, with B effective rational and $A = K_X + B$ $\mathbb{Q}$-Cartier and semiample. Choose a sufficiently divisible integer $k \geq 2$ such that kB is integral and kA is Cartier and globally generated, and a general section $s_0 \in H^0(X, \mathcal{O}_X(kA))$ with zero divisor G, such that*

$$\left(X, B + \frac{k-1}{k}G\right) \quad \textit{is klt.}$$

Put

$$U = X_{\mathrm{reg}} \setminus \mathrm{Supp}(B + G).$$

Let $q : V \to U$ be the finite étale $\boldsymbol{\mu}_k$-torsor of roots of the nowhere vanishing k-canonical form $s_0|_U$. All its connected components are included. Its tautological top form ω satisfies $\omega^k = q^* s_0$. For each $2 \leq j \leq k$ with jB integral, put $F_j = \mathcal{O}_X(jA)$ and define

$$\beta_j : F_j|_U \longrightarrow q_* \omega_V, \qquad \beta_j(u) = \frac{q^* u}{\omega^{j-1}}. \tag{3.5}$$

Then restriction, β_j, and the top holomorphic de Rham term induce an injection

$$\mathrm{per}_{j,0} : H^1(X, F_j) \hookrightarrow H^{n+1}(V, \mathbb{C}), \qquad n = \dim X. \tag{3.6}$$

Proof. We first justify the choice of s_0. Resolve (X, B) after the given compact Kähler modification, using projective blowups. The pullback linear system $|kA|$ is generated. A general member is smooth, transverse to the finitely many resolution strata, and contains no exceptional center. Adding its pullback with coefficient $(k-1)/k < 1$ to the crepant boundary gives an SNC divisor whose coefficients are all less than one. The asserted pair is therefore klt. The divisor G is reduced and shares no component with B. If the generated system is trivial, take $G = 0$. The same choice makes $(X, B + (j-1)G/k)$ klt for every $2 \leq j \leq k$.

Step 1: a compact Kähler model of the cover. Put $S_X = X \setminus U$. Further projective principalization on the chosen Kähler modification gives

$$\mu : T \longrightarrow X, \qquad S = (\mu^{-1}(S_X))_{\mathrm{red}} \text{ an SNC divisor}, \qquad T_U = \mu^{-1}(U),$$

where T is smooth compact Kähler. The map $\mu_U : T_U \to U$ is a proper modification of smooth manifolds.

The meromorphic k-canonical form $\mu^* s_0$ is holomorphic on T_U. Choose an effective integral divisor H supported on S which clears its poles, and put $L = \mathcal{O}_T(K_T + H)$. The monic root hypersurface of this section of $L^{\otimes k}$ is finite over T and compact inside $\mathbf{P}(\mathcal{O}_T \oplus L)$, disjoint from the infinity section. It is reduced: the monic root algebra is locally free over $\mathcal{O}_T$ and generically separable, so any nilpotent would be torsion and must vanish.

Let N_T be its normalization. There is a canonical identification

$$(N_T)|_{T_U} = T_U \times_U V. \tag{3.7}$$

To check it, write $s_0 = f\tau^k$ on a coordinate open in U, with f a unit and τ a volume form. On an open in T_U write $\mu^*\tau = J\eta$, where η is a volume form. The root equation is $y^k = (f \circ \mu)J^k$. Its normalization is $z^k = f \circ \mu$, via $y = Jz$: the latter cover is finite étale and normal, and the finite map agrees with the root equation where $J \neq 0$. These local identifications glue to (3.7).

Take an equivariant embedded resolution of the root hypersurface in $\mathbf{P}(\mathcal{O}_T \oplus L)$ and resolve its inverse image of S. Its smooth strict transform Z is compact Kähler, since the ambient

operations are projective blowups. It factors through N_T. Let $p : Z \to X$ be the composite and set

$$W = p^{-1}(U), \qquad E = (p^{-1}(S_X))_{\mathrm{red}} = Z \setminus W.$$

We choose the resolution so that E is SNC. Equation (3.7) gives a proper modification

$$h : W \longrightarrow V, \qquad p|_W = q \circ h. \tag{3.8}$$

It is the composite of the restricted resolution with the base change of μ_U, and is bimeromorphic on every component. Both W and V are smooth.

The cyclic action lifts to Z. Its tautological root, interpreted as a meromorphic canonical form, gives a meromorphic top form on Z restricting to $h^*\omega$ on W; denote it again by ω. Let χ be the character for which $\zeta^*\omega = \zeta\omega$.

Step 2: the character eigensheaf. The required sheaf identity is

$$\big(p_*\omega_Z(E)\big)_{\chi^{1-j}} = F_j. \tag{3.9}$$

Over U this identification is $h^*\beta_j$. Indeed, $h_*\omega_W = \omega_V$ for the smooth proper modification h, by the Hartogs argument used in Lemma 3.1. Consequently

$$(p_*\omega_Z(E))|_U = q_*h_*\omega_W = q_*\omega_V.$$

The character summands of the degree-k torsor are line bundles, and (3.5) identifies its χ^{1-j} summand with $F_j|_U$.

For the forward inclusion in (3.9), let u be a local section of F_j, and let $D_u \geq 0$ be its zero divisor as a section of this reflexive adjoint sheaf. It is $\mathbb{Q}$-Cartier. Put $\Theta_j = B + (j-1)G/k$. For a divisorial valuation Q over X and an extension R upstairs with ramification index e, the order of the meromorphic top form is

$$\operatorname{ord}_R \beta_j(u) = e\big(v_Q(D_u) + a(Q, X, \Theta_j) + 1\big) - 1. \tag{3.10}$$

The discrepancy convention is that of Section 2. To verify the formula, realize Q on a smooth model $\mu' : T' \to X$ and put $a_Q = \operatorname{coeff}_Q(K_{T'} - \mu'^*A)$. The pulled-back pluricanonical forms u and s_0 have orders

$$A_Q = v_Q(D_u) + ja_Q, \qquad C_Q = v_Q(G) + ka_Q$$

on T'. At a divisor of ramification index e, their further pullbacks have orders $eA_Q + j(e-1)$ and $eC_Q + k(e-1)$. Since $\omega^k = p^*s_0$, the order of ω is $eC_Q/k + e - 1$. Consequently

$$\begin{aligned} \operatorname{ord}_R \beta_j(u) &= eA_Q + j(e-1) - (j-1)(eC_Q/k + e - 1) \\ &= e\big(v_Q(D_u) + a_Q - (j-1)v_Q(G)/k + 1\big) - 1, \end{aligned}$$

which is (3.10).

To apply the calculation to an arbitrary prime of Z, use analytic flattening of the proper generically finite map [22]. After a modification of the base, the strict transform is proper and flat of relative dimension zero, hence finite. The prime then lies over a divisor on the modified base. Normalization and resolution reduce to the discrete valuation ring calculation above.

Since (X, Θ_j) is klt and D_u is effective $\mathbb{Q}$-Cartier, (3.10) is an integer greater than -1. It is therefore nonnegative at every prime of Z. Thus $\beta_j(u)$ extends holomorphically across all divisors, and hence over Z.

For the reverse inclusion, let σ be a local logarithmic top form of character χ^{1-j} on Z. Over W, descent by h and the torsor identification express it uniquely as $\beta_j(u)$. To extend u meromorphically, choose a nonzero local meromorphic canonical frame τ on X. The quotient

$$\frac{\sigma\omega^{j-1}}{(p^*\tau)^j} \tag{3.11}$$

is an invariant meromorphic function, hence descends meromorphically to the normal quotient $Z/\boldsymbol{\mu}_k$. The induced map $Z/\boldsymbol{\mu}_k \to X$ is proper and bimeromorphic: over a dense open subset of U, the map h is an equivariant isomorphism and $V/\boldsymbol{\mu}_k = U$. Meromorphic descent through this modification, followed by multiplication by τ^j, gives u. For modification descent, the proper image of the graph in $X \times \mathbf{P}^1$ is analytic and generically single-valued, hence defines the descended function.

At a generic boundary prime with coefficient $b \in (0,1)$, let d be the integer order of u relative to F_j. Logarithmic growth and (3.10) imply

$$e(d+1-b)-1 \geq -1, \qquad d \geq b-1 > -1.$$

Thus $d \geq 0$. At a generic component of the reduced divisor G the same inequality gives

$$d \geq \frac{j-1}{k} - 1 > -1,$$

again implying $d \geq 0$. At every other prime divisor the cover is generically étale and E is absent, so no pole occurs. Modifications of a normal space are isomorphisms at its generic codimension-one points. Reflexivity of F_j now proves the reverse inclusion in (3.9).

Step 3: the period map. By (3.9), the character projector identifies F_j with a direct summand of $p_*\omega_Z(E)$. The Leray edge map and (3.1) therefore give

$$H^1(X, F_j) \hookrightarrow H^1(Z, \Omega_Z^n(\log E)) \hookrightarrow H^{n+1}(W, \mathbb{C}). \tag{3.12}$$

The sheaf map restricts on W to $h^*\beta_j$. Naturality of the Leray edge map and de Rham comparison identifies this composite with

$$H^1(X, F_j) \xrightarrow{\mathrm{per}_{j,0}} H^{n+1}(V, \mathbb{C}) \xrightarrow{h^*} H^{n+1}(W, \mathbb{C}). \tag{3.13}$$

Since this composite is injective, $\mathrm{per}_{j,0}$ is injective. □

Remark 3.3 (The canonical character). The two period constructions use different open sets. For $j \geq 2$, the character bound along G identifies the eigensheaf with F_j. In degree one, the invariant logarithmic character allows a simple pole along G, so Lemma 3.1 instead uses $U = X_{\mathrm{reg}}$. This also gives the degree-one injection for an arbitrary klt central fiber with a compact Kähler modification.

4 Infinitesimal periods and analytic extension

We apply Levine's infinitesimal extension method [31, 32] to the central period maps of Section 3. First, a cohomological induction converts the period injections into lifting over Artinian thickenings. A finite-module argument then realizes these lifts analytically.

4.1 An Artinian period criterion

Lemma 4.1. *Let $g : \mathcal{Y} \to \Delta$ be proper and let $\mathcal{F}$ be a coherent sheaf flat over Δ. For $a \geq 0$, put*

$$R_a = \mathbb{C}[t]/(t^{a+1}), \qquad X_a = \mathcal{Y} \times_\Delta \operatorname{Spec} R_a, \qquad F_a = \mathcal{F}/t^{a+1}\mathcal{F},$$

and write $X = X_0$. Fix $r \geq 0$. Suppose there is a complex vector space H and, for $0 \leq a \leq r$, R_a-linear maps

$$\Phi_a : H^1(X_a, F_a) \longrightarrow H \otimes_{\mathbb{C}} R_a$$

compatible with truncation and with multiplication by t. If Φ_0 is injective, then every Φ_a is injective and every restriction

$$H^0(X_a, F_a) \longrightarrow H^0(X, F_0) \tag{4.1}$$

is surjective for $0 \leq a \leq r$.

Proof. Flatness gives

$$0 \longrightarrow F_{a-1} \xrightarrow{t} F_a \longrightarrow F_0 \longrightarrow 0.$$

The statement at $a = 0$ is the hypothesis together with the identity on H^0. Suppose it is known at $a - 1$. If a class $b \in H^1(X_{a-1}, F_{a-1})$ maps to zero under multiplication by t, then $t\Phi_{a-1}(b) = 0$. The map

$$H \otimes R_{a-1} \xrightarrow{t} H \otimes R_a$$

is injective. Thus $\Phi_{a-1}(b) = 0$, and $b = 0$ by induction. The long exact cohomology sequence gives the surjectivity in (4.1).

Next let $c \in H^1(X_a, F_a)$ satisfy $\Phi_a(c) = 0$. Its central restriction is zero by injectivity of Φ_0. Exactness therefore writes $c = tb$ for some $b \in H^1(X_{a-1}, F_{a-1})$. Compatibility and the same coefficient injection imply $\Phi_{a-1}(b) = 0$. Hence $b = 0$ and $c = 0$. This completes both parts of the induction. □

For a line bundle on a smooth proper family, the passage from finite-order lifting to constancy was proved in [31, Lemma 1.1]. The next two lemmas give a coherent-sheaf version of this passage.

Lemma 4.2 (Analytic realization)**.** *With the notation and properness and flatness assumptions of Lemma 4.1, suppose that $H^0(X_a, F_a) \to H^0(X, F_0)$ is surjective for every $a \geq 0$. Then every section of F_0 extends to a section of $\mathcal{F}$ over the inverse image of a neighborhood of 0. The lifts at the separate finite orders may be chosen independently.*

Proof. Put

$$R = \mathbb{C}\{t\}, \qquad M = (g_*\mathcal{F})_0, \qquad Q = (R^1 g_*\mathcal{F})_0.$$

Both are finite R-modules by Grauert's proper direct-image theorem [15]. The direct-image sequence for multiplication by t^a is

$$0 \longrightarrow M/t^a M \longrightarrow H^0(X_{a-1}, F_{a-1}) \xrightarrow{\partial_a} Q[t^a] \longrightarrow 0. \tag{4.2}$$

Here the sheaf supported on the Artinian central thickening has direct image stalk equal to the indicated cohomology group. The natural diagram from multiplication by t^a to multiplication by t has multiplication by t^{a-1} on its left term and the identity on its middle term. Its connecting maps give

$$\partial_1 \circ \operatorname{res} = t^{a-1}\partial_a. \tag{4.3}$$

The torsion submodule of the finite module Q is killed by t^b for some finite $b \geq 0$. Choose $a > b$ and lift the prescribed central section to F_{a-1}. Equations (4.2) and (4.3) show that its ∂_1 image is zero. It therefore lies in the image of M/tM. A representative in M is, by the definition of a direct-image stalk, an analytic section on the inverse image of a sufficiently small disk. This is the desired extension. A finite basis of the central space gives a common shrinking for all its sections. □

Lemma 4.3 (Extension and base change)**.** *Let $g : \mathcal{Y} \to \Delta$ be proper and let $\mathcal{F}$ be coherent and flat over Δ. If every section of $\mathcal{F}|_{X_0}$ extends near X_0, then $h^0(X_t, \mathcal{F}|_{X_t})$ is constant near 0. The direct image is locally free there and commutes with arbitrary holomorphic base change.*

Proof. The coherent sheaf $g_*\mathcal{F}$ is torsion-free on the disk, because multiplication by a nonzero base germ is injective on $\mathcal{F}$. It is therefore locally free. The direct-image sequence for multiplication by t injects its fiber at zero into $H^0(X_0, \mathcal{F}|_{X_0})$, and the extension hypothesis makes this map surjective. Its rank thus equals the central dimension. The corresponding injection at a nearby point gives dimension at least this rank, while upper semicontinuity gives dimension at most the central dimension. Equality follows near zero. Grauert's theorem [15] gives base change. □

4.2 Degree one

Proof of Theorem D. Let $\mathcal{F} = \mathcal{O}_{\mathcal{Y}}(K_{\mathcal{Y}/\Delta})$. Lemmas 2.1 and 2.2, with the usual canonical adjunction, give flatness and $F_0 = \omega_X$. Put $U = X_{\mathrm{reg}}$ and let $U_a = X_a|_U$. The morphism $U_a/\operatorname{Spec} R_a$ is smooth, and

$$F_a|_{U_a} = \omega_{U_a/R_a}.$$

For every a, the relative holomorphic Poincaré lemma gives the canonical quasi-isomorphism

$$\underline{R_a} \simeq \Omega^{\bullet}_{U_a/R_a}, \qquad \mathbb{H}^{n+1}(U_a, \Omega^{\bullet}_{U_a/R_a}) = H^{n+1}(U, \mathbb{C}) \otimes R_a. \tag{4.4}$$

Locally this is the holomorphic Poincaré lemma in relative coordinates with coefficients in R_a; the canonical inclusions of constant scalars glue to give the displayed quasi-isomorphism.

Restriction, the top term of the relative de Rham complex, and (4.4) define

$$\Phi_a : H^1(X_a, F_a) \longrightarrow H^{n+1}(U, \mathbb{C}) \otimes R_a.$$

They commute with multiplication by t and truncation, and their central map is injective by Lemma 3.1. Apply Lemma 4.1 to every finite truncation and then Lemma 4.2. □

4.3 Semiample adjoints

The proof uses an auxiliary pluricanonical degree to control every smaller admissible degree, following the argument that Levine credits to N. I. Shepherd-Barron [31, Corollary 1.8]. Lemma 3.2 supplies the required character period maps for reflexive adjoints.

Proof of Theorem C. In degree $m = 1$, admissibility and the coefficient bounds give $\mathcal{B} = B = 0$, so Theorem D applies. A zero-dimensional normal connected central fiber is a reduced point; its flat deformation is smooth near that point, and the assertion follows directly. Hence assume $m \geq 2$ and $n = \dim X > 0$.

Step 1: periods from an auxiliary lift. For $j = m$ and a sufficiently divisible $j = k$, set

$$\mathcal{F}_j = \mathcal{O}_{\mathcal{Y}}(j\mathcal{A}), \qquad F_{j,a} = \mathcal{F}_j/t^{a+1}\mathcal{F}_j.$$

Lemma 2.2 identifies $F_{j,0}$ with $\mathcal{O}_X(jA)$ and gives the exact sequences of thickenings. Choose $k \geq m$ such that $k\mathcal{A}$ is Cartier near the compact central fiber, $k\mathcal{B}$ is integral, and kA is globally generated. A common Cartier multiple exists near the whole central fiber by compactness, and a further multiple preserves generation on X.

Choose s_0 and G as in Lemma 3.2. Write

$$U = X_{\mathrm{reg}} \setminus \mathrm{Supp}(B+G), \qquad q : V \to U.$$

Given a lift

$$s_r \in H^0(X_r, F_{k,r}) \tag{4.5}$$

of s_0, use its compatible truncations s_a for $0 \leq a \leq r$. On $U_a = X_a|_U$, the morphism is smooth and the boundary is absent. Thus

$$F_{j,a}|_{U_a} = \omega_{U_a/R_a}^{\otimes j},$$

and s_a is a unit as a section of the kth canonical power. Its canonical root cover

$$q_a : V_a \longrightarrow U_a, \qquad \omega_a^k = q_a^* s_a$$

is finite étale and smooth over R_a. Its underlying topological space is V, and the covers and tautological forms reduce compatibly along this chosen finite chain.

By the same relative Poincaré calculation,

$$\mathbb{H}^{n+1}(V_a, \Omega^\bullet_{V_a/R_a}) = H^{n+1}(V, \mathbb{C}) \otimes R_a.$$

The maps

$$\beta_{j,a}(u) = \frac{q_a^* u}{\omega_a^{j-1}}$$

therefore define periods

$$\mathrm{per}_{j,a} : H^1(X_a, F_{j,a}) \longrightarrow H^{n+1}(V, \mathbb{C}) \otimes R_a, \qquad j = m, k. \tag{4.6}$$

The forms ω_a are truncations of ω_r, so these maps commute with truncation and multiplication by t.

Lemma 3.2 makes the central maps injective. Consequently Lemma 4.1 gives, for this finite chain,

$$\mathrm{per}_{j,a} \text{ injective}, \qquad H^0(X_a, F_{j,a}) \twoheadrightarrow H^0(X, F_{j,0}), \qquad 0 \leq a \leq r, j = m, k. \tag{4.7}$$

Thus a lift of s_0 to order r gives lifts of every section of $\mathcal{O}_X(mA)$ to that order. It remains to construct an auxiliary lift for every r; Lemma 4.2 will then give the analytic extension.

Step 2: the auxiliary obstruction. Suppose that s_r in (4.5) has been constructed. Let

$$\delta_{r+1}(s_0) \in H^1(X_r, F_{k,r})$$

be the connecting class in

$$0 \longrightarrow F_{k,r} \xrightarrow{t} F_{k,r+1} \longrightarrow F_{k,0} \longrightarrow 0. \tag{4.8}$$

Its vanishing gives an order-$r+1$ lift of the central section s_0. This is the inductive assertion: at each stage we require a lift of s_0 to the indicated order.

Levine's differential calculation [31, Lemmas 1.2–1.5] expresses this obstruction on U_r. Choose a Stein coordinate cover of the smooth locus near U, and transport sections to a

common chart when taking Čech differences. We use $(\check{\delta}a)_{ij} = a_i - a_j$. Write the relative transition functions and their Jacobian determinants as

$$z_i = g_{ij}(z_j, t), \qquad J_{ij} = \det\left(\frac{\partial g_{ij}}{\partial z_j}\right).$$

Differentiate the analytic transition functions before reducing modulo t^{r+1}. In the target coordinates, put

$$\rho_{ij} = \sum_{\alpha} \frac{\partial g_{ij}^{\alpha}}{\partial t} \frac{\partial}{\partial z_i^{\alpha}}. \tag{4.9}$$

Differentiating the coordinate cocycle identity shows that the resulting relative vector fields form a Čech cocycle on U_r.

Write the chosen section on U_r in canonical coordinates as $s_r = h_i(dz_i)^k$. Choose arbitrary local lifts $\widetilde{h}_i$ of its coefficients to order $r+1$. The gluing relation modulo t^{r+1} reads

$$h_i(g_{ij}(z_j, t), t) J_{ij}^k = h_j(z_j, t).$$

The chosen lifts therefore satisfy

$$\widetilde{h}_i(g_{ij}(z_j, t), t) J_{ij}^k - \widetilde{h}_j(z_j, t) = t^{r+1}\zeta_{ij} \pmod{t^{r+2}}.$$

Its differentiated discrepancy, modulo t^{r+1}, leaves the cocycle

$$D(s_r)_{ij} = \big(k \operatorname{div}(\rho_{ij}) h_i + \rho_{ij}(h_i)\big)(dz_i)^k. \tag{4.10}$$

With this Čech convention, its class is

$$[D(s_r)] = (r+1)\delta_{r+1}(s_0)|_{U_r}. \tag{4.11}$$

Indeed, division of the difference of the chosen lifts by t represents the connecting class in (4.8) by $t^r\zeta_{ij}$. Differentiation represents $(r+1)$ times this class. The local $\partial_t \widetilde{h}_i$ terms form a coboundary, leaving (4.10).

Step 3: vanishing of the period. Finite étaleness lifts ρ_{ij} to relative vector fields on V_r. Their contractions with the global tautological form,

$$\mu_{ij} = \iota_{\rho_{ij}} \omega_r,$$

form a Čech cocycle of relative $(n-1)$-forms. Since $q_r^* s_r = \omega_r^k$, Lie differentiation and Cartan's formula give

$$\begin{aligned} \beta_{k,r}\big(D(s_r)_{ij}\big) &= \frac{\mathcal{L}_{\rho_{ij}}(\omega_r^k)}{\omega_r^{k-1}} \\ &= k\, \mathcal{L}_{\rho_{ij}} \omega_r = k\, \mathrm{d}\mu_{ij}. \end{aligned} \tag{4.12}$$

The last expression is a boundary in the total Čech–de Rham complex, because the Čech differential of μ is zero. Hence (4.11) has zero period. Injectivity of $\mathrm{per}_{k,r}$ in (4.7), and characteristic zero, imply

$$\delta_{r+1}(s_0) = 0. \tag{4.13}$$

Starting with s_0, induction gives an auxiliary lift at every finite order. Step 1 lifts every section of $\mathcal{O}_X(mA)$ to each such order, and Lemma 4.2 realizes its analytic extension. □

Remark 4.4. The Kähler modification provides the central period injections. The relative obstruction calculation takes place on the smooth finite-order covers of U_r; injectivity then detects the global obstruction on X_r.

4.4 Canonical central fibers

Corollary 4.5. *Let $g : \mathcal{Y} \to \Delta$ be proper and flat. Assume that the total space is normal and canonical near the connected central fiber X, and that X is normal and canonical. Suppose X admits a compact Kähler modification and K_X is semiample. Then for every fixed $m \geq 1$, every section of $\omega_X^{[m]}$ extends near X. After shrinking so that the nearby fibers are normal and canonical,*

$$t \longmapsto h^0(X_t, \omega_{X_t}^{[m]})$$

is constant, and the corresponding direct image is locally free and satisfies holomorphic base change.

Proof. Canonical singularities are klt, and the canonical divisor is $\mathbb{Q}$-Cartier. The usual adjunction for a normal Cartier fiber identifies $K_{\mathcal{Y}/\Delta}|_X$ with K_X. Thus Theorem C, with boundary zero, extends every reflexive canonical power. The analytic deformation theorem for canonical singularities [28, Lemma 10 and Remark 10.3] permits a shrinking on which all fibers are normal and canonical. Lemma 2.2, applied at each such fiber, identifies the restriction of $\mathcal{O}_{\mathcal{Y}}(mK_{\mathcal{Y}/\Delta})$ with $\omega_{X_t}^{[m]}$. Lemma 4.3 now gives the remaining conclusions for the fixed degree m. □

In degree one, Theorem D gives the same extension conclusion without semiampleness.

5 Uniform neighborhoods and graded algebras

A generated multiple groups all adjoint powers into finitely many residue classes. Pushing these classes to relative projective space gives one neighborhood for all degrees and controls the multiplication maps. This is the finite-generation principle used in [28, Lemma 11 and paragraph 13]. We first treat global sections, then graded algebras and higher cohomology.

Lemma 5.1 (Uniformity from a generated multiple)**.** *Let $g : \mathcal{Y} \to \Delta$ be proper and flat. Suppose that coherent sheaves $\mathcal{F}_m$, $m \geq 0$, are flat over Δ, with $\mathcal{F}_0 = \mathcal{O}_{\mathcal{Y}}$. Suppose that, for an integer $k > 0$, $L = \mathcal{F}_k$ is invertible and there are identifications*

$$\mathcal{F}_{ak+r} = \mathcal{F}_r \otimes L^a, \qquad a \geq 0, \quad 0 \leq r < k,$$

compatible with fiber restriction. Assume that L_0 is globally generated and that every central section of every $\mathcal{F}_m$, $m \geq 1$, extends to a germ over the disk. Then there is a disk about 0 on which all the dimensions

$$h^0(Y_t, \mathcal{F}_m|_{Y_t}), \qquad m \geq 1,$$

are constant. On that disk $g_*\mathcal{F}_m$ is locally free and commutes with holomorphic base change for every $m \geq 1$.

Proof. Step 1: a generated multiple. Extend a finite basis of $H^0(Y_0, L_0)$ and shrink once. The cokernel of the resulting evaluation $\mathcal{O}_{\mathcal{Y}}^{N+1} \to L$ vanishes at every central point by Nakayama's lemma. Its support is closed; properness of g allows a further shrinking on which it is empty. Consequently these sections define a proper morphism

$$\phi : \mathcal{Y} \longrightarrow P := \mathbf{P}^N \times \Delta, \qquad L = \phi^*\mathcal{O}_P(1).$$

Properness follows from the closed graph and the properness of g.

Step 2: base change for the residue sheaves. For the finite set of residues $0 \le r < k$, put

$$E_r = \phi_* \mathcal{F}_r, \qquad Q_r = R^1 \phi_* \mathcal{F}_r.$$

They are coherent. Each E_r is torsion-free over the disk, hence flat. At the origin, the long exact sequence for multiplication by t gives

$$0 \longrightarrow E_r/tE_r \longrightarrow (\phi_0)_*(\mathcal{F}_r|_{Y_0}) \longrightarrow T_r \longrightarrow 0, \qquad T_r = Q_r[t]. \tag{5.1}$$

We prove $T_r = 0$ by extending sections in one sufficiently large degree $ak + r$ for each residue r.

Fix r and choose one sufficiently large integer a such that

$$H^1(\mathbf{P}^N, (E_r/tE_r)(a)) = 0, \qquad T_r(a) \text{ is globally generated}, \qquad ak + r > 0.$$

Serre vanishing and generation, applied through GAGA [46], give such an a. Every section of $\mathcal{F}_{ak+r}|_{Y_0}$ extends by hypothesis. The projection formula makes its extension a section of $E_r(a)$ near the central projective space. Therefore

$$H^0(\mathbf{P}^N, (E_r/tE_r)(a)) \longrightarrow H^0(\mathbf{P}^N, (\phi_0)_*(\mathcal{F}_r|_{Y_0})(a))$$

is surjective. Taking cohomology in (5.1) and using the chosen H^1 vanishing gives $H^0(T_r(a)) = 0$. Global generation forces $T_r = 0$.

The equality $Q_r[t] = 0$ says that Q_r is flat over the disk at every point of P_0. Relative flatness of a coherent analytic sheaf is open [12]. The projection $P \to \Delta$ is proper, so we may shrink once to make all the finitely many Q_r flat over the whole new disk. At any point c of this disk, the multiplication-by-$t - c$ sequence now gives

$$E_r|_{P_c} \simeq (\phi_c)_*(\mathcal{F}_r|_{Y_c}). \tag{5.2}$$

Step 3: uniform regularity. Choose one nonnegative integer a_0 for which all the central sheaves $E_r|_{P_0}$ are a_0-regular, that is,

$$H^i(\mathbf{P}^N, E_r|_{P_0}(a_0 - i)) = 0 \quad (1 \le i \le N,\ 0 \le r < k).$$

These are finitely many conditions. Upper semicontinuity and one further shrinking make them hold at every nearby c. Castelnuovo–Mumford regularity [37, Lecture 14] then gives

$$H^i(\mathbf{P}^N, E_r|_{P_c}(a)) = 0 \quad (i > 0,\ a \ge a_0,\ 0 \le r < k) \tag{5.3}$$

simultaneously on this disk. For $N = 0$ there is no higher cohomology and the same conclusion is immediate.

Flatness of E_r makes the Euler characteristic of each fixed fiberwise twist constant on the connected disk. Together with (5.3), this makes $h^0(\mathbf{P}^N, E_r|_{P_c}(a))$ independent of c for every $a \ge a_0$. By (5.2) and the projection formula these are exactly the dimensions in degrees $ak+r$. Only finitely many positive degrees remain. Lemma 4.3 handles them, followed by a finite intersection of neighborhoods. Grauert's theorem gives the asserted local freeness and base change on the already fixed disk, for every degree. □

Corollary 5.2 (Semiample canonical central fibers)**.** *Let $g : \mathcal{Y} \to \Delta$ be proper and flat. Suppose that the connected central fiber Y_0 is normal and canonical, admits a compact Kähler modification, and has semiample canonical divisor. After shrinking once, the total space and all fibers are canonical, $K_{\mathcal{Y}/\Delta}$ is relatively semiample, and*

$$h^0(Y_t, \omega_{Y_t}^{[m]}) = h^0(Y_0, \omega_{Y_0}^{[m]}) \qquad (t \in \Delta,\ m \ge 1).$$

For every $m \ge 1$, the sheaf $g_*\mathcal{O}_{\mathcal{Y}}(mK_{\mathcal{Y}/\Delta})$ is locally free and commutes with holomorphic base change on this disk. Every central section extends over it.

Proof. By [28, Lemma 10 and Remark 10.3], shrink so that the total space and every fiber are normal canonical. The nearby fibers are connected by upper semicontinuity of $h^0(\mathcal{O})$. Take a common relative Cartier index and a multiple k for which $\omega_{Y_0}^{[k]}$ is generated. With

$$\mathcal{F}_m = \mathcal{O}_{\mathcal{Y}}(mK_{\mathcal{Y}/\Delta}), \qquad L = \mathcal{O}_{\mathcal{Y}}(kK_{\mathcal{Y}/\Delta}),$$

the restrictions and flatness follow from Lemma 2.2. Tensoring the reflexive sheaves $\mathcal{F}_r$ with L^a gives the residue identities. Central extension in every degree is Corollary 4.5. Apply Lemma 5.1. Its first step makes $L = \mathcal{O}_{\mathcal{Y}}(kK_{\mathcal{Y}/\Delta})$ relatively generated, which proves relative semiampleness. Finally, the locally free direct image on the Stein disk extends every vector in its central fiber to a section over the disk. □

Corollary 5.3 (Uniform logarithmic endpoint)**.** *Under the assumptions of Theorem C, let $d > 0$ be the least integer for which both $d\mathcal{B}$ and dB are integral. After shrinking, write B_t for the boundary given by restriction and adjunction on the normal fiber Y_t. Then there is a disk about 0 on which, for every fiber,*

$$h^0(Y_t, \mathcal{O}_{Y_t}(m(K_{Y_t} + B_t))) = h^0(X, \mathcal{O}_X(m(K_X + B))) \qquad (m \in d\mathbb{Z}_{>0}).$$

The sheaves $g_*\mathcal{O}_{\mathcal{Y}}(m\mathcal{A})$ are locally free and commute with holomorphic base change for all such m. Moreover, a positive Cartier multiple of $K_{\mathcal{Y}/\Delta} + \mathcal{B}$ is generated over the disk.

Proof. Normality of the fibers is open, so we first shrink around the normal central fiber. Since $\mathcal{Y}$ is klt, $K_{\mathcal{Y}/\Delta}$ is $\mathbb{Q}$-Cartier; hence so is $\mathcal{B} = \mathcal{A} - K_{\mathcal{Y}/\Delta}$. Its restriction defines B_t and the usual adjunction gives $\mathcal{A}|_{Y_t} = K_{Y_t} + B_t$. At a general codimension-one point of a normal fiber the morphism is smooth, so dB_t is integral. Lemma 2.2 therefore applies to every fiber.

Use $\mathcal{F}_j = \mathcal{O}_{\mathcal{Y}}(jd\mathcal{A})$. Theorem C extends every central section of every $\mathcal{F}_j$. Choose k such that $kd\mathcal{A}$ is Cartier and kdA is globally generated. The required residue identities follow by tensoring with the invertible $\mathcal{F}_k$. Lemma 5.1 and its generation argument prove the assertions. □

5.1 The graded algebra

Proposition 5.4 (The graded algebra at a semiample endpoint)**.** *In the setting of Lemma 5.1, assume in addition that $\mathcal{Y}$ is normal, the central fiber is normal and connected, and the sheaves $\mathcal{F}_m$ form a commutative graded $\mathcal{O}_{\mathcal{Y}}$-algebra with degree-zero part $\mathcal{O}_{\mathcal{Y}}$. Assume that the residue identifications in that lemma are induced by its multiplication. After shrinking to one disk U, the algebra*

$$\mathcal{R} = \bigoplus_{m \geq 0} g_*\mathcal{F}_m$$

is finitely generated and flat over $\mathcal{O}_U$, and every graded piece is locally free. In the base-change notation of Section 2, the natural map for every holomorphic $u : T \to U$ is an isomorphism of graded algebras:

$$u^*\mathcal{R} \simeq \bigoplus_{m \geq 0} (g_T)_*(p^*\mathcal{F}_m). \tag{5.4}$$

The relative analytic space

$$\mathcal{C} = \operatorname{Proj}_U \mathcal{R}$$

is normal, projective and flat over U, and its formation commutes with these base changes.

Proof. Use the disk and notation of Lemma 5.1. In particular, $L = \mathcal{F}_k$ is relatively generated, $\phi : \mathcal{Y} \to P = \mathbf{P}^N \times U$ satisfies $L = \phi^*\mathcal{O}_P(1)$, and $E_r = \phi_*\mathcal{F}_r$ for $0 \le r < k$. After shrinking, $g_*\mathcal{O}_\mathcal{Y} = \mathcal{O}_U$: upper semicontinuity gives $h^0(Y_t, \mathcal{O}_{Y_t}) = 1$, and Grauert's theorem identifies the unit map with an isomorphism.

Finite generation and base change. The uniform Castelnuovo–Mumford regularity used in that lemma also gives, for every $t \in U$, $r < k$ and $a \ge a_0$, the surjection

$$H^0(\mathbf{P}^N, \mathcal{O}(1)) \otimes H^0(P_t, E_r|_{P_t}(a)) \longrightarrow H^0(P_t, E_r|_{P_t}(a+1));$$

see [37, Lecture 14]. The restriction isomorphisms (5.2), the projection formula, and Lemma 5.1 identify this with the fiber of

$$\mathcal{O}_U^{N+1} \otimes \mathcal{R}_{ak+r} \longrightarrow \mathcal{R}_{(a+1)k+r}.$$

Its cokernel vanishes by Nakayama's lemma. Thus the finitely many coherent sheaves $\mathcal{R}_{ak+r}$ with $0 \le a \le a_0$ and $0 \le r < k$ generate $\mathcal{R}$ as a module over $\mathrm{Sym}_{\mathcal{O}_U}(\mathcal{O}_U^{N+1})$, whose generators have degree k. This proves finite generation after one further shrinking. The algebra is flat over U, being a direct sum of locally free graded pieces. Degreewise holomorphic base change is Lemma 5.1; its naturality with respect to multiplication proves (5.4).

The projective model. Take the Stein factorization

$$\mathcal{Y} \xrightarrow{h} \mathcal{C}' \xrightarrow{v} P$$

of ϕ. The normality of $\mathcal{Y}$ implies the normality of $\mathcal{C}'$. Set $A = v^*\mathcal{O}_P(1)$, which is ample over U because v is finite. Since $h_*\mathcal{O}_\mathcal{Y} = \mathcal{O}_{\mathcal{C}'}$ and $L = h^*A$, the kth Veronese algebra is

$$\mathcal{R}^{(k)} = \bigoplus_{a \ge 0} g_*L^a = \bigoplus_{a \ge 0} (g_{\mathcal{C}'})_*A^a.$$

The relative Proj of the section algebra of an ample line bundle recovers its space. Relative Proj is unchanged by a Veronese subalgebra, so $\mathcal{C} = \mathcal{C}'$. This also proves projectivity. After shrinking, the normal total space $\mathcal{Y}$ is irreducible and dominates U; the same holds for $\mathcal{C}'$. Its local rings are therefore torsion-free over the local discrete valuation rings of U, and $\mathcal{C} \to U$ is flat.

Finally, after taking a sufficiently divisible Veronese, a finite homogeneous presentation realizes $\mathcal{C}$ as a closed analytic subspace of a projective bundle over U. Pulling back its homogeneous equations proves that relative analytic Proj commutes with arbitrary holomorphic base change. Together with (5.4), this identifies every base-changed projective model. □

5.2 Higher cohomological degrees

Lemma 5.5 (Uniformity in all cohomological degrees)**.** *Let $g : \mathcal{Y} \to \Delta$ be proper and flat, and let $\mathcal{F}_m$, $m \ge 0$, be coherent sheaves flat over Δ, with $\mathcal{F}_0 = \mathcal{O}_\mathcal{Y}$. Suppose that $L = \mathcal{F}_k$ is invertible for some $k > 0$, that L_0 is globally generated, and that*

$$\mathcal{F}_{ak+r} \simeq \mathcal{F}_r \otimes L^a \qquad (a \ge 0,\ 0 \le r < k),$$

compatibly with fiber restriction. Assume that, for every fixed $m \ge 1$, all the functions $t \mapsto h^i(Y_t, \mathcal{F}_m|_{Y_t})$ are constant near 0. Then there is a disk about 0 on which all these dimensions are constant simultaneously for $m \ge 1$ and $i \ge 0$. Every $R^ig_*\mathcal{F}_m$, with $m \ge 1$, is locally free there and commutes with holomorphic base change.

Proof. Grauert's theorem in degree k extends a basis of $H^0(Y_0, L_0)$. As in Lemma 5.1, its evaluation generates L after shrinking and defines

$$\phi : \mathcal{Y} \to P = \mathbf{P}^N \times \Delta, \qquad L = \phi^* \mathcal{O}_P(1), \qquad g = \pi\phi.$$

For $0 \le r < k$, set $Q_{r,j} = R^j \phi_* \mathcal{F}_r$. Proper coherence gives a finite collection of coherent sheaves: the terms with $j > \dim Y_0$ vanish.

Step 1: flatness of the higher direct images. Choose one integer $a > 0$, and shrink the disk, so that

$$R^p \pi_* Q_{r,j}(a) = 0 \quad (p > 0), \qquad Q_{r,j}[t](a) \text{ is globally generated on } P_0 \tag{5.5}$$

for every r, j. To make this choice, consider the t-power torsion subsheaf of each $Q_{r,j}$ near the compact fiber P_0. Coherence and compactness give a common annihilating power of t, so this torsion has a finite filtration with coherent quotients on P_0. The torsion-free quotient is flat at P_0 and, after shrinking, flat over the disk [12]. Apply Serre vanishing to the filtration quotients and to the central fiber of the flat quotient. Upper semicontinuity and Grauert's theorem give the first part of (5.5) for one sufficiently large a. Serre generation for the finitely many $Q_{r,j}[t]$ gives the second part.

The projection formula and Leray now identify

$$\pi_* Q_{r,j}(a) \simeq R^j g_* \mathcal{F}_{ak+r}.$$

The right side is locally free at the origin by the fixed-degree hypothesis and Grauert's theorem. The t-torsion of the left side is $H^0(P_0, Q_{r,j}[t](a))$. It is zero; generation in (5.5) therefore gives $Q_{r,j}[t] = 0$. Thus all $Q_{r,j}$ are flat at P_0. Openness of flatness and properness permit one shrinking on which they are all flat.

Step 2: fiberwise base change. For each c in this disk, the direct-image sequence for multiplication by $t - c$ gives

$$0 \longrightarrow Q_{r,j}/(t-c)Q_{r,j} \longrightarrow R^j(\phi_c)_*(\mathcal{F}_r|_{Y_c}) \longrightarrow Q_{r,j+1}[t-c] \longrightarrow 0.$$

The last term vanishes by flatness. Hence

$$Q_{r,j}|_{P_c} \simeq R^j(\phi_c)_*(\mathcal{F}_r|_{Y_c}) \tag{5.6}$$

for every r, j and c.

Step 3: uniform regularity. Choose a common regularity bound a_0 for the finitely many central sheaves $Q_{r,j}|_{P_0}$. Their flatness and upper semicontinuity make the same bound valid for every nearby fiber. Castelnuovo–Mumford regularity [37, Lecture 14] gives

$$H^p(P_c, Q_{r,j}|_{P_c}(a)) = 0 \qquad (p > 0,\ a \ge a_0).$$

Consequently, fiberwise Leray and (5.6) identify

$$H^j(Y_c, \mathcal{F}_{ak+r}|_{Y_c}) \simeq H^0(P_c, Q_{r,j}|_{P_c}(a)) \qquad (a \ge a_0).$$

The dimensions on the right equal Euler characteristics, which are constant because $Q_{r,j}$ is flat. This treats every large degree on one disk. Only finitely many positive degrees remain; their fixed-degree neighborhoods have a common intersection. Grauert's theorem proves local freeness and base change there. □

6 Canonical negative steps

The central program must start on the given canonical fiber. We use the steps of [28, Definition 7]: diagrams

$$X \xrightarrow{f} Z \xleftarrow{q} X^+$$

of projective bimeromorphic contractions such that $-K_X$ is f-ample, q is small, and K_{X^+} is q-ample. We call such a diagram a *canonical step*; it includes the mixed steps of [28, Definition 7]. The negative contraction may contract several divisors. Throughout this section the spaces are normal and irreducible.

6.1 The divisorial canonical algebra

We first recall the divisorial-algebra construction [30, 3.52 and Lemma 6.2] over a neighborhood of a prescribed compact subset of the target. It supplies the positive side of a canonical step; compare [28, Definition 7].

Lemma 6.1. *Let Z be a normal complex analytic space whose divisorial canonical algebra*

$$\mathcal{R}_Z = \bigoplus_{m\geq 0} \omega_Z^{[m]}$$

is locally finitely generated, and let $K \subset Z$ be compact. There are an open neighborhood U of K and an integer $r > 0$ such that, for

$$q : Y = \mathrm{Proj}_U(\mathcal{R}_Z|_U) \longrightarrow U,$$

Y is normal, q is small, projective and bimeromorphic, and $\omega_Y^{[r]}$ is invertible and q-ample. In particular, K_Y is $\mathbb{Q}$-Cartier and q-ample. If Z is compact, one may take $U = Z$.

Proof. Local finite generation supplies, around each point of K, an open set on which a sufficiently divisible Veronese is generated in degree one. Choose finitely many such sets V_i covering K, and let r be a common multiple of their Veronese degrees. On $U = \bigcup_i V_i$, the tautological sheaf $L = \mathcal{O}_Y(r)$ is invertible and relatively ample. Thus q is projective.

Normality can be checked over a connected Stein open set $V \subset U$. Cartan's theorem A gives a section of ω_V nonzero at a chosen smooth point; it supplies a local meromorphic canonical trivialization. Write its canonical Weil divisor as $\sum_P k_P P$. In the meromorphic function field with an added variable T, the divisorial algebra is the intersection of the polynomial ring with the valuation rings defined by the Gauss valuations

$$w_P\left(\sum_m a_m T^m\right) = \min_m\big(v_P(a_m) + mk_P\big).$$

The inequalities $w_P \geq 0$ express precisely the codimension-one regularity of each homogeneous coefficient. This intersection is integrally closed. The degree-zero rings after homogeneous localization are consequently normal, so the relative analytic Proj is normal. The same construction in the excellent analytic local rings gives normal analytic Proj charts. Over the open set where the canonical sheaf is invertible, this Proj is the original space. Thus it is bimeromorphic.

The relative-Proj comparison holds locally for all sufficiently large degrees. A finite covering of K, followed by a shrinking of U, gives one bound n_0 such that

$$q_* L^n = \omega_U^{[rn]} \qquad (n \geq n_0). \tag{6.1}$$

Suppose that E is a q-exceptional prime divisor. Viewing all the following sheaves in the same meromorphic rank-one sheaf, normality and reflexivity give

$$q_*L^n \subseteq q_*(L^n \otimes \mathcal{O}_Y(E)) \subseteq (q_*L^n)^{**} = \omega_U^{[rn]} = q_*L^n \qquad (n \geq n_0).$$

The second inclusion holds because the sheaves agree off $q(E)$, which has codimension at least two. Hence all inclusions are equalities. Relative Serre generation makes $L^n \otimes \mathcal{O}_Y(E)$ generated over a neighborhood of any fixed point of $q(E)$ for sufficiently large n. Its evaluation map, however, now factors through the proper subsheaf $L^n \subsetneq L^n \otimes \mathcal{O}_Y(E)$ at the generic point of E. This contradiction proves smallness.

The equality $L = \omega_Y^{[r]}$ holds on the isomorphism locus of q. Both sheaves are reflexive, and the complement has codimension at least two, so it holds everywhere. Thus K_Y is $\mathbb{Q}$-Cartier and q-ample, with the single index r on Y. □

Lemma 6.2. *Let $f : X \to Z$ be a projective bimeromorphic contraction of normal analytic spaces. Suppose X is canonical and $-K_X$ is f-ample. Fix a compact subset $K \subset Z$. There is an open neighborhood U of K on which the following assertions hold. Replace Z by U and X by $f^{-1}(U)$, retaining the notation $f : X \to Z$ for the restricted morphism. Then*

$$f_*\omega_X^{[m]} = \omega_Z^{[m]} \qquad (m \geq 0). \tag{6.2}$$

The relative Proj of this algebra completes f to a canonical step $X \dashrightarrow X^+$. The space X^+ is canonical. On a common projective resolution $p : W \to X$, $a : W \to X^+$, there is a $\mathbb{Q}$-linear equivalence of holomorphic $\mathbb{Q}$-line bundles

$$p^*K_X \sim_{\mathbb{Q}} a^*K_{X^+} + E_f, \qquad E_f \geq 0, \qquad a_*E_f = 0. \tag{6.3}$$

If f is nontrivial, $E_f \neq 0$; its coefficient is positive at the strict transform of every f-exceptional prime of X. The space Z has rational singularities. If the original target is compact Kähler, no restriction is necessary, and X^+ is compact Kähler.

Proof. For (6.2), work over a connected Stein neighborhood $V \subset Z$. Choose a meromorphic canonical frame on V as in the preceding proof, and pull it back to $f^{-1}(V)$. A reflexively regular m-canonical form on Z defines a $\mathbb{Q}$-Cartier divisor $D_X = \operatorname{div}(\varphi) + mK_X$ on X, with $f_*D_X \geq 0$ and $-D_X$ relatively nef. Exceptional negativity gives $D_X \geq 0$. Conversely, a form regular on X is regular at the generic points of the prime divisors of Z, and hence belongs to its reflexive canonical power. This proves (6.2).

Fujino's analytic form of canonical-algebra finite generation [14, Theorem 1.18], extending [3, Corollary 1.1.2], applies to $(X, 0)$ and f. Apply Lemma 6.1 to the prescribed compact subset K. After restriction to the resulting neighborhood and with the retained notation, it gives a small projective $q : X^+ \to Z$ with a single Cartier multiple of K_{X^+} that is q-ample. On the common resolution, subtract the two pulled-back canonical divisors defined by each local frame on Z. A change of frame adds the same pulled-back principal divisor on both sides, so the differences glue to a global rational discrepancy divisor E_f. Thus (6.3) is a global $\mathbb{Q}$-linear equivalence. Since q is small, every prime of X^+ comes from a prime of X; hence E_f is a-exceptional. On an a-vertical curve C, the image $p(C)$ is a point or an f-vertical curve, so

$$E_f \cdot C = p^*K_X \cdot C \leq 0.$$

Negativity proves $E_f \geq 0$.

If $E_f = 0$, lift a nontrivial f-vertical curve to a curve on W. Its degree against p^*K_X is negative, whereas its degree against $a^*K_{X^+}$ is nonnegative, a contradiction. If an f-exceptional

prime had zero coefficient, a general a-vertical curve through a general point of its strict transform would not be contained in $\operatorname{Supp} E_f$. Its E_f-degree would be nonnegative. Its image on X is nontrivial, since p is an isomorphism at that general point, so the same equality instead gives negative degree. This proves strictness at every contracted original prime.

The comparison increases discrepancies. A valuation exceptional over X has nonnegative discrepancy there. A prime of X contracted by the step has discrepancy zero on X and acquires a positive coefficient in the comparison. These exhaust the exceptional valuations over X^+, which is therefore canonical. Relative Kawamata–Viehweg vanishing, in the analytic form [14, Theorem 5.2] with boundary and integral divisor both zero, gives $R^i f_* \mathcal{O}_X = 0$ for $i > 0$. A resolution of X and Leray show that Z has rational singularities. If the original target is compact Kähler, apply the construction to the entire original target, so that no restriction is needed. A complex space projective over a compact Kähler space is compact Kähler, giving the last assertion. □

6.2 Supporting one negative analytic ray

We use the generalized-pair terminology of [20, 42]. The nef datum is represented by a nef class on a fixed smooth Kähler carrier; its trace on a working model is the birational pushforward of that class. All adjoint classes below are Bott–Chern classes. The pushforward of a big carrier class is modified big, and a modified-big Bott–Chern class on a Kähler space is big [20, Definition 2.8]. The *analytic cone* is the closed cone of classes generated by positive closed currents of bidimension $(1,1)$.

Lemma 6.3. *Let T be a canonical compact Kähler space. Let $A = \{K_T\} + \beta_T$ be a gklt adjoint with globally nef carrier data and big trace β_T. Suppose R is an A-negative extremal ray of the full analytic cone and $K_T \cdot R < 0$. Then R has a projective contraction $f : T \to Z$ with normal compact Kähler target and $-K_T$ relatively ample. If f is birational, it has the canonical replacement of Lemma 6.2 over all of Z.*

Proof. We use the supporting-class argument of [42, proof of Lemma 3.6]. The generalized cone theorem of Hacon–Li–Xie [17, Corollary 3.6], building on Hacon–Păun [19, Proposition 3.1 and Corollary 3.2], expresses the analytic cone as its A-nonnegative part and finitely many A-negative rational rays; see also [20, Theorem 1.3]. We claim that R is the null face of a nef supporting class ℓ. Normalize the analytic cone by a Kähler class. The cone generated by its A-nonnegative part and the finitely many other negative rays is closed: its normalized slice is the compact convex hull of the corresponding closed slices. It does not contain the normalized point of R. Separate that point from this cone and add the unique positive multiple of the normalizing Kähler class which makes the value on R zero. The result is strictly positive on the complementary cone and zero on R, as required by cone duality.

For $b \gg 0$, the class $\omega_R = b\ell - A$ is Kähler. Indeed, $-A$ is positive near R on the compact normalized slice, and ℓ has positive minimum on the complementary compact set. Add the pullback of ω_R to the fixed nef carrier data. The generalized discrepancies do not change; the new datum is nef and its trace remains big. Theorem 1.4 of [20] applies to the nef adjoint $A + \omega_R = b\ell$. It supplies a Moishezon contraction to a normal compact Kähler space with

$$b\ell = f^*\gamma, \qquad \gamma \text{ Kähler on } Z.$$

Its null analytic face is the single ray R.

Since K_T is $\mathbb{Q}$-Cartier and $K_T \cdot R < 0$, [42, Lemma 3.16] gives a relatively ample Cartier multiple of $-K_T$, so f is projective. In this setting, positivity on the compact cone slice makes $f^*\gamma - s\{K_T\}$ Kähler for small $s > 0$; a local base potential can be absorbed in a metric on

a fixed Cartier multiple of $-K_T$, proving relative ampleness of this multiple. The birational assertion now follows from Lemma 6.2, applied with $K = Z$. □

For scaling, we also need to transport Bott–Chern classes through the step. The degree along the contracted ray determines the exceptional correction; the remaining class descends to the base.

Lemma 6.4. *Consider a birational step obtained in Lemma 6.3. Every Bott–Chern class θ on T has a linear transform θ^+ on T^+. With the notation of (6.3),*

$$p^*\theta - a^*\theta^+ = c(\theta)\{E_f\}, \qquad c(\theta) = \frac{\theta \cdot R}{K_T \cdot R}. \tag{6.4}$$

The class θ^+ represents the birational pushforward of θ. In particular, $\{K_T\}^+ = \{K_{T^+}\}$.

Proof. The source and contraction base have rational singularities. Relative vanishing gives $R^i f_* \mathcal{O}_T = 0$ for $i > 0$. Every contracted curve has analytic class in R. Consequently $\theta - c(\theta)\{K_T\}$ has degree zero on all contracted curves. Degree-zero descent [20, Lemma 2.39] gives a unique class η on Z with

$$\theta - c(\theta)\{K_T\} = f^*\eta.$$

Set $\theta^+ = q^*\eta + c(\theta)\{K_{T^+}\}$ and use (6.3). This proves linearity and (6.4). Pushing its exceptional difference to the new model proves the trace assertion. □

7 Finite programs and divisorial negative parts

We first construct canonical programs with scaling. A termination criterion, expressed in terms of divisorial negative parts, then produces the finite program used in Section 8. The same criterion will also apply to the logarithmic program of Section 10.

7.1 Divisorial negative parts

For a pseudoeffective Bott–Chern class α on a smooth compact Kähler manifold, let

$$\nu_D(\alpha) = \lim_{\varepsilon \downarrow 0} \inf\{\nu_D(T) : T \in \alpha,\ T \geq -\varepsilon\omega\}, \qquad N(\alpha) = \sum_D \nu_D(\alpha) D,$$

where ω is a fixed Kähler form. This definition is independent of ω, and $N(\alpha)$ is an effective divisor with finite support [4, Sections 2–3 and Theorem 3.12]. For a holomorphic $\mathbb{Q}$-line bundle J, write $N(J) = N(\{J\})$. For a holomorphic $\mathbb{Q}$-line bundle on a normal compact Kähler space, a divisorial multiplicity is computed after pullback to a smooth Kähler resolution carrying that divisor. On a given model, $N(J)$ denotes the sum over the prime divisors of that model. For effective divisors N, B, the notation $N \wedge B = 0$ means that they have no common prime component.

We use homogeneity, monotonicity under addition of a nef class, and exceptional translation: if $q : W \to Y$ is a modification, α is pseudoeffective, and $E \geq 0$ is q-exceptional, then

$$N(q^*\alpha + \{E\}) = N(q^*\alpha) + E. \tag{7.1}$$

The normal-space formulation and its compatibility with higher resolutions are recorded in [42, Lemmas 2.2–2.4].

7.2 Canonical scaling

A *positive truncation* of a scaling program consists of the steps whose scaling thresholds are at least a fixed number $\varepsilon > 0$. The scaling parameter starts at 1; the initial Kähler class is chosen so that the adjoint at 1 is Kähler.

Lemma 7.1. *Let X be a canonical compact Kähler space. Choose a sufficiently large Kähler class H with $\{K_X\} + H$ Kähler. There is a K_X-negative program with scaling of H, possibly infinite, whose birational maps are the canonical steps of Section 6, and all working spaces are canonical and compact Kähler. It stops when the canonical class is nef or upon reaching a projective K-negative fiber-type contraction.*

Write $\alpha_i(s) = \{K_{X_i}\} + sH_i$ for the linear traces. Every positive truncation of the program is finite. If the program is infinite, its thresholds tend to zero. At a wall $s_i > 0$, the scaled pullbacks agree. On a common resolution $p_i : W_i \to X_i$, $a_i : W_i \to X_{i+1}$,

$$p_i^*\alpha_i(s) - a_i^*\alpha_{i+1}(s) = \left(1 - \frac{s}{s_i}\right)\{E_i\}, \qquad E_i \geq 0, \quad E_i \neq 0. \tag{7.2}$$

If K_X is not pseudoeffective, this program is finite and ends with a projective K-negative fiber-type contraction.

Proof. Start with $X_0 = X$, $H_0 = H$, and upper parameter $u = 1$. Suppose inductively that a finite prefix has reached X_i, that $\alpha_i(u)$ is nef, and that for $0 \leq s \leq u$ its comparison with the original scaled class is effective and exceptional over X_i. On a common smooth carrier, the nef datum remains the pullback of sH from X. Its trace on X_i is exactly sH_i by Lemma 6.4. Since the pullback of H is nef and big, this trace is modified big for $s > 0$. It is also a Bott–Chern class on the Kähler space X_i, so it is big [20, Definition 2.8]. On the common carrier, the effective comparison subtracts an effective divisor from the original generalized boundary. Thus $(X_i, 0 + s\mathbf{H})$ is gklt for $0 \leq s \leq u$. Its ordinary boundary is zero, since all preceding steps are nonextracting.

If K_{X_i} is nef, stop. Otherwise let

$$\tau = \min\{s \in [0, u] : \alpha_i(s) \text{ is nef}\} > 0, \qquad 0 < t_0 < \tau.$$

Apply the generalized cone theorem [17, Corollary 3.6] to the gklt big-trace adjoint $A = \alpha_i(t_0)$. Its negative cone has finitely many extremal rays. On the A-nonnegative part, all $\alpha_i(s)$ with $t_0 \leq s \leq u$ are nonnegative, being convex combinations of A and $\alpha_i(u)$. Thus τ is determined by the finitely many negative-ray inequalities. One such ray R satisfies

$$\alpha_i(\tau) \cdot R = 0, \qquad \alpha_i(t_0) \cdot R < 0.$$

It follows that $H_i \cdot R > 0$ and $K_{X_i} \cdot R = -\tau H_i \cdot R < 0$. Lemma 6.3 constructs the contraction of this single ray, using a new Kähler supporting direction on X_i. A fiber-type contraction ends the program. In the birational case, write its canonical replacement as $X_i \xrightarrow{f_i} Z_i \xleftarrow{q_i} X_{i+1}$.

The wall annihilates R, so degree-zero descent and Lemma 6.4 give

$$\alpha_i(\tau) = f_i^*\delta, \qquad \alpha_{i+1}(\tau) = q_i^*\delta.$$

The class δ is nef: nefness descends under a surjective map of normal compact Kähler spaces, by the general pullback criterion recalled in [20, Section 2.4]. Therefore the new wall is nef. The unscaled canonical comparison now gives (7.2), with $s_i = \tau$. For $s \leq \tau$ it preserves the effective-comparison and gklt assertions used above, so the construction can be repeated. The finiteness argument below also bounds the number of steps at any fixed positive threshold.

Fix $\varepsilon > 0$ and a resolution $\nu : W \to X$ of the original model. The compact polytope

$$\{(0, s\nu^* H) : \varepsilon \le s \le 1\}$$

has globally nef carrier data, gklt original pairs, and big traces. The finite marked weak-model theorem [20, Theorem 5.15(2)] applies in the category of Definition 2.15 of that paper. The nef wall and the composed effective comparison make every working model with threshold at least ε a marked weak model in that theorem. A marked model cannot recur: every nontrivial step has a nonzero effective unscaled canonical comparison, these comparisons add after pullback, and recurrence would give identical canonical discrepancies. There are therefore only finitely many steps with threshold at least ε. In particular, an infinite program has thresholds tending to zero.

If K_X is not pseudoeffective, let $\epsilon > 0$ be its pseudoeffective threshold in the direction H. Every nef working wall, together with its effective exceptional comparison, makes the original scaled class pseudoeffective. Thus every threshold is at least ϵ. Positive-truncation finiteness makes the program finite. A nef endpoint would make the unscaled original canonical class pseudoeffective by its effective comparison, a contradiction. The final contraction is therefore of fiber type. □

7.3 A termination criterion

The next criterion abstracts the affine comparison argument of [42, proof of Theorem 3.13]. First, a finite prefix removes all divisorial negative parts on the working model. A new scaling direction then gives a finite continuation to a nef model.

Proposition 7.2 (Termination from one nef interval)**.** *Let $\mathscr{C}$ be a class of normal compact Kähler models X equipped with holomorphic $\mathbb{Q}$-line bundles J, closed under the chosen negative steps. Suppose the following properties hold for every pseudoeffective member (X, J) of $\mathscr{C}$.*

(i) *For every sufficiently large Kähler class H, a J-negative program with scaling of H either reaches a nef model or continues indefinitely. Its maps are nonextracting, its transforms J_i are pseudoeffective, and every positive truncation is finite. The thresholds are positive and nonincreasing. Each scaled class $\{J_i\} + sH_i$ is nef between its incoming and outgoing thresholds; the classes H_i transform linearly, and the scaled pullbacks agree at each wall s_i. On a common resolution of each step,*

$$p_i^* J_i \sim_{\mathbb{Q}} q_i^* J_{i+1} + E_i, \qquad E_i \ge 0, \quad q_{i*} E_i = 0, \quad E_i \ne 0,$$

with positive coefficient at every prime contracted from X_i.

(ii) *If $N(J_T) = 0$ on a working model T, then for each sufficiently large Kähler class A there are a compact Kähler space V small bimeromorphic with T, transformed classes $\{J_V\}, A_V$, and $\delta > 0$ such that $\{J_V\} + sA_V$ is nef for $0 \le s \le \delta$. The two pullback differences on a common smooth resolution are represented by real divisors exceptional over both models.*

Then every pseudoeffective member (X, J) admits a finite J-negative program starting on X and ending at a nef model. If, on a resolution, each J_i has a decomposition $r^* J_i \sim_{\mathbb{Q}} P_i + N(r^* J_i)$ with P_i semiample, the endpoint is semiample.

Proof. Scaled comparisons. Wall agreement and linearity give

$$p_i^*(\{J_i\} + sH_i) - q_i^*(\{J_{i+1}\} + sH_{i+1}) = \left(1 - \frac{s}{s_i}\right)\{E_i\}. \tag{7.3}$$

On a nef working interval, compose these formulas. All previous thresholds are at least the parameter, so the resulting comparison is effective and exceptional over the working model. By (7.1), it is the full negative part of the original scaled class on the common resolution.

Removing divisorial negative parts. For the unscaled comparison, pseudoeffectivity and (7.1) place each contracted original prime in $\operatorname{Supp} N(J)$. This support is finite and nonextraction prevents a prime from reappearing. Thus only finitely many steps contract divisors.

Suppose the program is infinite. Its positive thresholds tend to zero, by positive-truncation finiteness. Choose a model T after the last divisorial step and after a finite equal-threshold block, with a nonempty nef working interval. For a parameter λ_0 in this interval, put $Q = \{J_T\} + \lambda_0 H_T$, which is nef. All later maps are small. For parameters λ in the later nef working intervals, the exceptional comparison has zero coefficient at each prime of T. Hence the minimal multiplicities of $\{J_T\} + \lambda H_T$ at those primes vanish. Such parameters tend to zero.

To pass to the limit, write

$$\{J_T\} + \lambda H_T = \left(1 - \frac{\lambda}{\lambda_0}\right)\{J_T\} + \frac{\lambda}{\lambda_0} Q. \tag{7.4}$$

On a fixed smooth Kähler resolution $r : W \to T$, choose a Kähler class Ω and $c > 0$ with $c\Omega - r^*Q$ nef. Then

$$\nu_D(r^*\{J_T\} + c\eta\Omega) \le \nu_D(r^*\{J_T\} + \eta r^*Q) \le \nu_D(r^*\{J_T\})$$

for every prime D on W. By the definition of minimal multiplicity, the outer terms have the same limit as $\eta \downarrow 0$. Homogeneity and (7.4) therefore give convergence to the multiplicities of J_T. It follows that $N(J_T) = 0$ on T.

A finite continuation. Restart scaling on T with a sufficiently large Kähler class A. The preceding strict comparison excludes divisorial steps. Choose V, δ as in (ii). On a common smooth resolution $p : W \to T$, $q : W \to V$, write

$$p^*(\{J_T\} + sA) = q^*(\{J_V\} + sA_V) + \{F(s)\}, \qquad 0 \le s \le \delta. \tag{7.5}$$

The divisor $F(s)$ is affine and exceptional over both models. Its negative is p-nef, so exceptional negativity gives $F(s) \ge 0$. Equation (7.1) now yields

$$N\big(p^*(\{J_T\} + sA)\big) = F(s) \qquad (0 \le s \le \delta). \tag{7.6}$$

This equality persists on higher resolutions.

If the restarted program were infinite, group the finitely many steps at each positive threshold into a block. Its effective unscaled comparisons add to a nonzero effective divisor, and (7.3) still holds for the block. The resulting distinct walls would tend to zero. Choose a wall $s_i \in (0, \delta)$ whose two adjacent working intervals also lie in $(0, \delta)$, and resolve the corresponding finite prefix with T and V. The nef comparisons on the two intervals compute the negative part of the same original class. At a prime having coefficient $e_i > 0$ in the block comparison, the two affine slopes differ by e_i/s_i. They cannot both agree with the single affine coefficient in (7.6). Thus the restarted program terminates. Joining it to the finite initial prefix gives the asserted program from X.

Semiampleness. At a nef endpoint, the good divisorial decomposition has zero negative part on a resolution. The pulled-back holomorphic $\mathbb{Q}$-line bundle is consequently semiample. Clear its Cartier index; normality and the projection formula identify its sections with those on the endpoint. Generation upstairs then implies generation downstairs. □

7.4 The finite canonical program

Proposition 7.3. *Let X be a normal compact Kähler space with canonical singularities. There is a finite program of canonical negative steps starting on X, with canonical compact Kähler working spaces. If K_X is pseudoeffective, its endpoint has semiample canonical divisor. Otherwise it ends with a projective K-negative fiber-type contraction.*

Proof. The non-pseudoeffective case is Lemma 7.1. In the pseudoeffective case, Proposition A.1 gives on a smooth compact Kähler modification $r : W \to X$ a $\mathbb{Q}$-linear equivalence of holomorphic $\mathbb{Q}$-line bundles

$$r^*K_X \sim_{\mathbb{Q}} P + N(r^*K_X), \qquad P \text{ semiample.} \tag{7.7}$$

Thus some pluricanonical power has a nonzero section. Canonical bimeromorphic invariance (Lemma 2.4) preserves this property on every working model. It excludes a negative fiber-type contraction: after raising the section to a Cartier-index-clearing power, its nonzero restriction to a general positive-dimensional projective fiber would lie in an antiample line. Every contraction target is compact, so Lemma 6.2 applies over the entire target. Together with Lemma 7.1, this verifies condition (i) of Proposition 7.2 for the class of canonical working models.

For condition (ii), apply Proposition A.2 to the klt pair $(T, 0)$. Proposition 7.2 now gives a finite program and, by (7.7), a semiample endpoint. □

8 Canonical invariance in analytic families

We follow Kollár's construction [28, paragraph 8]: extend the negative contraction and recover the positive side from its relative canonical algebra. Fujino's analytic finite-generation theorem [14, Theorem 1.18] supplies that algebra. Pluricanonical residue identifies its central quotient and hence the central positive model.

8.1 Deformation of canonical steps

Lemma 8.1 (Deformation of a canonical MMP step)**.** *Let $g : \mathcal{X} \to \Delta$ be proper and flat with normal connected canonical central fiber. Suppose that*

$$X_0 \xrightarrow{f_0} Z_0 \xleftarrow{q_0} Y_0$$

is a diagram of projective bimeromorphic contractions of normal compact complex spaces, with $-K_{X_0}$ relatively ample for f_0, q_0 small, and K_{Y_0} relatively ample for q_0. Assume Y_0 is canonical. After shrinking the disk there is a relative diagram

$$\mathcal{X} \xrightarrow{f} \mathcal{Z} \xleftarrow{q} \mathcal{Y}$$

having this central diagram, with $-K_{\mathcal{X}/\Delta}$ f-ample and $K_{\mathcal{Y}/\Delta}$ q-ample. The families are proper and flat, and the nearby fibers of $\mathcal{X}$ and $\mathcal{Y}$, as well as their total spaces, are canonical. The induced correspondences $X_t \dashrightarrow Y_t$ are bimeromorphic on every nearby fiber.

Proof. Extending the contraction. By [28, Lemma 10 and Remark 10.3], canonical singularities of the compact central fiber give, after shrinking, normal canonical total space, a $\mathbb{Q}$-Cartier relative canonical divisor, and canonical nearby fibers. Relative Kawamata–Viehweg vanishing [14, Theorem 5.2], applied to the projective K-negative contraction with $D = 0$, gives

$R^1(f_0)_*\mathcal{O}_{X_0} = 0$. The contraction-extension theorem [29, Proposition 11.4], in the form stated in [16, Theorem 4.7], extends f_0 to a contraction $f : \mathcal{X} \to \mathcal{Z}$ over a smaller disk, with proper flat target and central fiber Z_0. The identity $f_*\mathcal{O}_\mathcal{X} = \mathcal{O}_\mathcal{Z}$ gives a normal target. Its parameter is a nonzerodivisor, so it is flat over the disk; the parameter sequence and the central R^1 vanishing identify its central structure sheaf with $\mathcal{O}_{Z_0}$.

Take a relative Cartier multiple of $-K_{\mathcal{X}/\Delta}$. Its central restriction is f_0-ample. Openness of relative ampleness [16, Proposition 3.5] and compactness make this line f-ample after shrinking. Thus f is projective. It is bimeromorphic because its source and target have the same dimension and $f_*\mathcal{O}_\mathcal{X} = \mathcal{O}_\mathcal{Z}$.

The positive model and its central fiber. Apply Lemma 6.2 to f with the compact subset $Z_0 \subset \mathcal{Z}$. It gives the canonical replacement over an open neighborhood U of Z_0. Write $\pi : \mathcal{Z} \to \Delta$ for the proper target family. The set $\pi(\mathcal{Z} \setminus U)$ is closed and does not contain 0. Choose a smaller disk Δ_ε disjoint from this set, so that $\pi^{-1}(\Delta_\varepsilon) \subset U$. Restrict the diagram to this disk and retain the notation Δ for it. Denote the resulting canonical replacement by $q : \mathcal{Y} \to \mathcal{Z}$. The relative canonical algebra is locally finitely generated; $\mathcal{Y}$ is canonical, q is small, and $K_{\mathcal{Y}/\Delta}$ is q-ample. For the residue calculation, put

$$\mathscr{R} = \bigoplus_{m\geq 0} \omega_\mathcal{Z}^{[m]}(mZ_0). \tag{8.1}$$

Twisting by the Cartier divisor Z_0 does not change the Proj.

Choose a projective resolution $\tau : \mathcal{W} \to \mathcal{X}$ such that the strict transform W_0 of X_0 is smooth. Put $h = f\tau$. The smooth divisor W_0 is nonexceptional for the projective bimeromorphic map h. Nakayama's theorem [40, VI.3.8], in the precise analytic form of [28, Theorem 9], gives

$$h_*\omega_\mathcal{W}^{\otimes m}(mW_0) \twoheadrightarrow (h|_{W_0})_*\omega_{W_0}^{\otimes m}. \tag{8.2}$$

Canonicality of X_0 and the negative-side algebra identity identify the right side with $\omega_{Z_0}^{[m]}$. Codimension-one regularity identifies the left side with a subsheaf of $\omega_\mathcal{Z}^{[m]}(mZ_0)$. The flat morphism $\mathcal{Z} \to \Delta$ is smooth along $(Z_0)_{\mathrm{reg}}$. Pluricanonical residue is therefore defined there and extends to $\omega_{Z_0}^{[m]}$ by reflexivity. It agrees with the residue in (8.2). We obtain a surjection

$$\omega_\mathcal{Z}^{[m]}(mZ_0) \twoheadrightarrow \omega_{Z_0}^{[m]}.$$

Its kernel is exactly $t\omega_\mathcal{Z}^{[m]}(mZ_0)$. Indeed, a section of zero residue is divisible by t at the generic point of Z_0. At every other codimension-one point, t is a unit. The quotient by t consequently remains in the same reflexive sheaf by normality. Conversely, a multiple of t has zero residue. The restrictions are multiplicative, hence

$$\mathscr{R}/t\mathscr{R} \simeq \bigoplus_{m\geq 0} \omega_{Z_0}^{[m]}. \tag{8.3}$$

Relative Proj commutes with the quotient by t. Smallness of q_0 and q_0-ampleness of K_{Y_0} identify the Proj of the right-hand algebra in (8.3) with Y_0. Thus the actual central fiber of $\mathcal{Y}$ is the prescribed Y_0. The parameter is a nonzerodivisor on the normal bimeromorphic total space, so $\mathcal{Y} \to \Delta$ is flat; it is proper by construction. Canonicality of its nearby fibers follows from [28, Lemma 10] after shrinking.

Fiberwise bimeromorphicity. Both central arrows are isomorphisms over a common nonempty open subset of Z_0. Lemma 2.5 extends this common isomorphism locus to a total open set. Openness of $\mathcal{X} \to \Delta$ makes this set meet every sufficiently nearby fiber. Upper semicontinuity

of $h^0(\mathcal{O})$ makes the nearby normal fibers X_t and Y_t connected, hence irreducible. Their identified nonempty open subsets therefore give the asserted bimeromorphic correspondences. □

Lemma 8.2 (Deformation of the final negative fiber contraction)**.** *Let $\mathcal{M} \to \Delta$ be proper and flat, with connected canonical central fiber, and let $h_0 : M_0 \to Z_0$ be a projective contraction such that $\dim Z_0 < \dim M_0$ and $-K_{M_0}$ is h_0-ample. After shrinking, all reflexive plurigenera of all fibers of $\mathcal{M}$ are zero, on one disk independent of the degree.*

Proof. Canonical openness gives a canonical total space and canonical nearby fibers after shrinking. Upper semicontinuity of $h^0(\mathcal{O}_{M_t})$ makes these fibers connected. Relative vanishing [14, Theorem 5.2] gives $R^1(h_0)_*\mathcal{O}_{M_0} = 0$, and [29, Proposition 11.4] extends the contraction. This is the zero-boundary case of [16, Proposition 4.9]. The target is proper over the disk, and its fiber dimension is at most $\dim Z_0$ near zero. One common Cartier index q can be chosen for $K_{\mathcal{M}/\Delta}$. Openness of relative ampleness makes

$$A = \mathcal{O}_{\mathcal{M}}(-qK_{\mathcal{M}/\Delta})$$

relatively ample for the extended contraction. Since $\dim Z_t \le \dim Z_0 < \dim M_t$, the contraction has positive-dimensional fibers on a dense open subset of each nearby M_t.

Suppose $s \in H^0(M_t, \omega_{M_t}^{[m]})$ is nonzero. Its q-th reflexive power is a nonzero section of A_t^{-m}. Choose a positive-dimensional projective fiber meeting its nonvanishing locus, and then an integral curve in that fiber through a nonvanishing point. The restricted section is nonzero, whereas A_t^{-m} has strictly negative degree on that curve. This is impossible. Since A and the neighborhood were fixed before choosing m, the vanishing holds in every degree on that neighborhood. □

8.2 Families over a disk

Proposition 8.3 (The disk case)**.** *The assertions of Theorem A hold when the base is a disk.*

Proof. Reduction to the endpoint. Shrink using [28, Lemma 10 and Remark 10.3]. The total space and all fibers are normal canonical, and the relative canonical divisor is $\mathbb{Q}$-Cartier. Upper semicontinuity of $h^0(\mathcal{O}_{X_t})$ makes the nearby fibers connected, hence irreducible. The reflexive sheaves

$$\mathcal{F}_m = \mathcal{O}_{\mathcal{X}}(mK_{\mathcal{X}/\Delta})$$

are flat over the disk and their actual fiber restrictions are $\omega_{X_t}^{[m]}$ by Lemma 2.2.

Proposition 7.3 gives a finite birational program

$$X_0 = X_0^0 \dashrightarrow X_0^1 \dashrightarrow \cdots \dashrightarrow X_0^N = M_0$$

whose endpoint either has semiample canonical divisor or carries a projective K-negative fiber-type contraction. All central working spaces are canonical and compact Kähler. Lemma 8.1 extends the birational list over one smaller disk. Applying Lemma 2.4 with zero boundary to every fiber gives

$$P_m(X_t) = P_m(M_t) \qquad (t \in \Delta,\ m \ge 1). \tag{8.4}$$

In the semiample case, M_0 has a compact Kähler resolution, so Corollary 5.2 makes every $P_m(M_t)$ constant on one disk. In the fiber-type case, Lemma 8.2 makes all these plurigenera zero on one disk. Thus in both cases, simultaneously for every $m \ge 1$,

$$P_m(X_t) = P_m(M_t) = P_m(M_0) = P_m(X_0).$$

Direct images and section extension. Apply Grauert's theorem to the flat coherent sheaf $\mathcal{F}_m$ for each m on the already fixed disk. Each $f_*\mathcal{F}_m$ is locally free and has holomorphic base change. Surjectivity on the central fiber gives extension of every central pluricanonical section. Since the disk is Stein, Cartan's theorem A extends every such vector to a global section of its direct-image bundle. □

Remark 8.4 (Projective central fibers). For a projective central fiber, the finite program can also be obtained from the nonfactorial termination theorem of Hashizume–Hu [21, Theorem 1.7]. If K_{X_0} is pseudoeffective, [42, Proposition 2.10] gives a good minimal model. The theorem of Hashizume–Hu supplies a terminating negative MMP beginning on the given X_0, without a factoriality assumption. Its endpoint is good by [50, Lemma 2.10]. The mixed steps described in [50, Remark 2.15] are precisely the diagrams of [28, Definition 7]. Hence [28, Theorem 2] extends the entire finite program. The uniform endpoint theorem and canonical bimeromorphic invariance give the disk assertion of Proposition 8.3. If K_{X_0} is not pseudoeffective, [21, Theorem 1.7] instead supplies a Mori program, and Lemma 8.2 handles all degrees together.

8.3 General analytic bases

We use the following disk criterion for the full sequence of plurigenera; compare [28, paragraph 13].

Lemma 8.5 (Detection by disks)**.** *Let S be a complex analytic space, possibly nonreduced, let Λ be a set, and let $u : |S| \to \Lambda$ be a function. If $u \circ \gamma$ is constant for every holomorphic map $\gamma : \Delta \to S$, then u is locally constant.*

Proof. The assertion is local on S. Reduction preserves the underlying topological space, and every map from a disk factors through S_{red}. Take a proper surjective resolution $\pi : T \to S_{\mathrm{red}}$ [1, Theorem 13.3], using the disjoint union of resolutions of the local irreducible components when necessary. Any two points in a coordinate polydisk of T can be joined by a finite chain of coordinate disks. The hypothesis therefore makes $u \circ \pi$ constant on every such polydisk, hence locally constant. For $s_0 \in S$, the set

$$W = \{t \in T : u(\pi(t)) = u(s_0)\}$$

is open and contains $\pi^{-1}(s_0)$. Since π is proper,

$$U = S \setminus \pi(T \setminus W)$$

is an open neighborhood of s_0. Surjectivity gives $u(s) = u(s_0)$ for every $s \in U$. □

Proof of Theorem A. Proposition 8.3 gives the assertions over a disk. For the numerical assertion over an arbitrary base, replace S by S_{red} and pull back the family. Properness, flatness and all point fibers are preserved.

Nakayama's stability theorem [40, VI.5.2], in the analytic form of [28, Lemma 10], and properness give a neighborhood of 0 on which every fiber is canonical. Canonical singularities are rational. By Bingener's deformation theorem [2, Theorem 6.3], in the form of [27, Corollary 3.3.6], all these fibers are Kähler after a further restriction. Openness of flat maps and upper semicontinuity of $h^0(X_s, \mathcal{O}_{X_s})$ also make the fibers nonempty and connected. Denote the resulting neighborhood by V and set

$$\Phi : V \longrightarrow \mathbb{Z}_{\geq 0}^{\mathbb{Z}_{>0}}, \qquad \Phi(s) = \big(P_m(X_s)\big)_{m \geq 1}.$$

For a holomorphic map $\gamma : \Delta \to V$, the pullback family is proper and flat, and every fiber satisfies the central-fiber hypotheses of Proposition 8.3. Applying that proposition at

each point of the disk shows that $\Phi \circ \gamma$ is locally constant, hence constant because the disk is connected. Lemma 8.5 implies that Φ is locally constant, giving one neighborhood of 0 for all degrees. Reduction preserves both point fibers and underlying open sets, so this is also the required neighborhood on the original base. □

9 Canonical algebras and deformation consequences

9.1 Canonical and adjoint algebras

The semiample family constructed in the proof of Proposition 8.3 determines the relative canonical algebra. Birational comparison identifies its sections and their products with those of the original family.

Proof of Theorem E. Under hypothesis (a), suppose first that $\kappa(X_0) = -\infty$. Theorem A gives $P_m(X_t) = 0$ for every $m > 0$ on one disk U. The fibers are connected and normal, so $f_*\mathcal{O}_{\mathcal{X}} = \mathcal{O}_U$. Thus $\mathcal{R}(f) = \mathcal{O}_U$ is concentrated in degree zero and $\operatorname{Proj}_U \mathcal{R}(f)$ is empty.

In the remaining cases $\kappa(X_0) \geq 0$. Under hypothesis (a), retain the proper flat family $g : \mathcal{Y} \to U$ obtained at the semiample endpoint in the proof of Proposition 8.3. Its total space and fibers are canonical, and the correspondence with $\mathcal{X}$ is bimeromorphic on every fiber. Under hypothesis (b), take $\mathcal{Y} = \mathcal{X}$ and apply Corollary 5.2.

Choose a Cartier multiple $kK_{\mathcal{Y}/U}$ whose central restriction is globally generated. Lemma 5.1 extends a generating set and makes this multiple relatively generated on one disk. Hence $K_{\mathcal{Y}/U}$ is relatively semiample, proving (iii). The sheaves $\mathcal{O}_{\mathcal{Y}}(mK_{\mathcal{Y}/U})$ form a graded algebra under multiplication of meromorphic pluricanonical forms. Lemma 2.2 identifies their fiber restrictions and gives their flatness. Tensoring by $\mathcal{O}_{\mathcal{Y}}(kK_{\mathcal{Y}/U})$ gives the residue-class identities required by Proposition 5.4. That proposition proves (i) and (ii) for g.

Under (a), Lemma 2.4 on the total spaces identifies the direct-image canonical algebras of $\mathcal{X}$ and $\mathcal{Y}$. The identifications preserve multiplication because they identify meromorphic pluricanonical forms. Applying the same lemma on each fiber gives the stated fiber algebras. The disk assertion of Theorem A gives base change for the original sheaves $\mathcal{F}_m$, and naturality of multiplication gives the graded base-change isomorphism. This proves (i) and (ii) for f. □

For $\kappa(X_0) = \dim X_0$, the relative Proj is the relative canonical model. For $0 \leq \kappa(X_0) < \dim X_0$, it is the relative Iitaka model, with fibers of dimension $\kappa(X_0)$.

Corollary 9.1 (The admissible adjoint algebra)**.** *Under the assumptions of Theorem C, let $d > 0$ be the least integer for which $d\mathcal{B}$ and dB are integral. After restriction to a disk U about 0, $\mathcal{A}$ is relatively semiample and*

$$\mathcal{R}_d(g, \mathcal{A}) := \bigoplus_{j \geq 0} g_*\mathcal{O}_{\mathcal{Y}}(jd\mathcal{A})$$

is a finitely generated flat $\mathcal{O}_U$-algebra with locally free graded pieces. The natural graded base-change map (5.4) is an isomorphism for every holomorphic base change. Its relative analytic Proj is normal, projective and flat over U, and its formation also commutes with base change.

Proof. Corollary 5.3 gives relative semiampleness and a common disk for all admissible degrees. The sheaves $\mathcal{F}_j = \mathcal{O}_{\mathcal{Y}}(jd\mathcal{A})$ form a graded algebra, and their residue-class identities respect multiplication. Proposition 5.4 applies and proves the remaining assertions. □

9.2 Higher pluricanonical cohomology

Nakayama's theorem gives local freeness of pluricanonical higher direct images for a proper family over a disk with klt total space, relatively semiample canonical divisor and class-$\mathcal{C}$ fiber components [39, Corollary 3.15]; see [33, Proposition 4.6]. For smooth families, the following result assumes semiampleness and class $\mathcal{C}$ on the central fiber.

The fixed-degree argument below combines Levine's cyclic-cover construction with the eigenspace argument credited there to Shepherd-Barron [31, Corollary 1.8]. Lemma 5.5 then makes the neighborhood uniform in both the canonical and cohomological degrees.

Corollary 9.2 (Higher pluricanonical cohomology)**.** *Let $f : \mathcal{X} \to \Delta$ be a proper holomorphic submersion of complex manifolds with connected fibers. Suppose that X_0 belongs to Fujiki's class $\mathcal{C}$ and that K_{X_0} is semiample. There is a disk U about 0 such that*

$$h^i(X_t, K_{X_t}^{\otimes m}) = h^i(X_0, K_{X_0}^{\otimes m}) \qquad (t \in U,\ i \geq 0,\ m \in \mathbb{Z}).$$

For every $i \geq 0$ and $m \in \mathbb{Z}$, the sheaf $R^i f_* K_{\mathcal{X}/U}^{\otimes m}$ is locally free on U and commutes with holomorphic base change.

Proof. Corollary 5.2 gives a relatively generated power of $K_{\mathcal{X}/\Delta}$ on a disk about 0. Set $n = \dim X_0$; cohomology in degrees $i > n$ vanishes. We first prove constancy for a fixed $m \geq 1$ and all $0 \leq i \leq n$. Choose a sufficiently divisible $q \geq m$ such that $K_{X_0}^{\otimes q}$ is globally generated. A general section has a smooth, possibly empty, zero divisor. Extend it by Corollary 4.5 and shrink the disk so that its divisor is smooth over the base. The associated cyclic cover

$$p : \mathcal{Z} \longrightarrow \mathcal{X}, \qquad p_* \mathcal{O}_{\mathcal{Z}} = \bigoplus_{j=0}^{q-1} K_{\mathcal{X}/\Delta}^{\otimes(-j)}$$

is smooth and proper over this disk.

Each connected component of Z_0 belongs to class $\mathcal{C}$. Indeed, pull the cover back to a compact Kähler modification $\mu : T \to X_0$. The pullback is a closed subspace of the projective completion of $\mu^* K_{X_0}$, which is a compact Kähler manifold. Resolving the components that dominate T gives compact Kähler modifications of the components of Z_0; every component of the finite cover $Z_0 \to X_0$ dominates X_0. By Fujiki's degeneration theorem [13, Proposition 1.6], the Frölicher spectral sequence degenerates at E_1 on each component of Z_0. In the smooth proper family $\mathcal{Z} \to \Delta$, Betti numbers are constant and Hodge numbers are upper semicontinuous. For each total degree, the Frölicher inequality and equality at 0 force every Hodge number to be constant near 0. In particular, each $h^\ell(Z_t, \mathcal{O}_{Z_t})$ is constant.

The cyclic-cover decomposition gives

$$h^\ell(Z_t, \mathcal{O}_{Z_t}) = \sum_{j=0}^{q-1} h^\ell(X_t, K_{X_t}^{\otimes(-j)}).$$

Each summand is upper semicontinuous, and their sum is constant; hence each summand is constant. Taking $j = m-1$ and $\ell = \dim X_0 - i$, Serre duality proves the fixed-degree assertion.

Apply Lemma 5.5 to $\mathcal{F}_m = K_{\mathcal{X}/\Delta}^{\otimes m}$ and to the relatively generated canonical power chosen at the beginning. It gives one disk for all $m \geq 1$ and all i, together with local freeness and base change. Serre duality identifies degrees m and $1-m$, giving constancy for $m \leq 0$ on the same disk. Grauert's theorem gives the remaining direct-image assertions. □

Corollary 9.3 (Nef central adjoints)**.** *Assume the hypotheses of Theorem C, with X compact Kähler and $A = K_X + B$ nef in place of semiample. Then Corollaries 5.3 and 9.1 apply. If the family is smooth and $\mathcal{B} = 0$, the conclusions of Corollary 9.2 also hold.*

Proof. Proposition A.1 makes the nef adjoint A semiample. Apply the indicated corollaries. □

In particular, Theorem A establishes the constancy of plurigenera requested in [6, Question 1.6]; it applies to the compact Kähler central fiber in that question in every Kodaira dimension.

9.3 Numerical consequences

Hacon–Xie prove invariance of canonical pseudoeffectivity and uniruledness for families with Kähler total space [20, Corollary 7.8], resolving [16, Conjecture 1.4]. The following corollary gives these conclusions for families of compact Kähler canonical fibers.

Corollary 9.4 (Numerical and birational invariants)**.** *Let $f : \mathcal{X} \to S$ be a proper flat morphism of complex analytic spaces with S connected and not necessarily reduced. Suppose that all fibers are connected n-dimensional compact Kähler spaces with canonical singularities. Their Kodaira dimension and canonical volume are constant. The loci*

$$\{s \in S : K_{X_s} \text{ is pseudoeffective}\}, \qquad \{s \in S : X_s \text{ is uniruled}\}$$

are each either empty or equal to S.

Proof. The case $n = 0$ is immediate; assume $n > 0$. Theorem A and connectedness of S make each $P_m(X_s)$ constant. This sequence determines the Kodaira dimension and the canonical volume

$$\operatorname{vol}(K_{X_s}) := \limsup_{m\to\infty} \frac{n!\, P_m(X_s)}{m^n}.$$

For each fiber, Proposition A.1 gives

$$K_{X_s} \text{ pseudoeffective} \quad \Longleftrightarrow \quad P_m(X_s) > 0 \text{ for some } m > 0.$$

For the forward implication, take a common multiple of the semiample positive part and the effective rational negative part in the good decomposition. A nonzero section of the resulting pullback descends by the projection formula. The reverse implication follows from the effective pluricanonical divisor. Pseudoeffectivity is therefore constant.

Choose a compact Kähler resolution $W_s \to X_s$. Canonical birational invariance gives $\kappa(W_s) = \kappa(X_s)$. The same nonvanishing argument on W_s identifies $\kappa(W_s) = -\infty$ with non-pseudoeffectivity of K_{W_s}. Ou's criterion [43, Theorem 1.1] identifies this condition with uniruledness of W_s, and uniruledness is birationally invariant. It is consequently constant on the fibers X_s as well. □

10 Finite logarithmic programs

For logarithmic invariance, we use divisorial contractions and flips between globally strongly $\mathbb{Q}$-factorial terminal pairs. The scaling construction of [42, Lemmas 3.6–3.7] and Proposition 7.2 produce a finite program starting on the prepared central pair.

10.1 Ordinary logarithmic steps

A normal space is *globally strongly $\mathbb{Q}$-factorial* if every global coherent rank-one reflexive sheaf has an invertible reflexive power. In particular every global Weil divisor is $\mathbb{Q}$-Cartier.

This property holds on a smooth space. An *ordinary negative step* for $J = K_X + B$ means a projective divisorial contraction or a projective small contraction and its flip, with normal contraction base and connected fibers, such that $-J$ is relatively ample on the contracting side and the transformed adjoint is relatively ample on the positive side. Boundaries are pushed forward or strictly transformed.

Lemma 10.1 (Ordinary logarithmic scaling)**.** *Let (X, B) be a globally strongly $\mathbb{Q}$-factorial compact Kähler klt pair with effective rational boundary, and put $J = K_X+B$. For a sufficiently large Kähler class A, there is an ordinary J-negative program with scaling of A. Its working spaces remain compact Kähler and globally strongly $\mathbb{Q}$-factorial. Its maps are nonextracting and every positive truncation is finite. If J is pseudoeffective, the program either reaches a nef adjoint or has positive thresholds tending to zero.*

On a common resolution of a finite prefix, the unscaled comparison is

$$p^*J \sim_{\mathbb{Q}} q^*J_i + E_i, \qquad E_i \geq 0, \quad q_*E_i = 0. \tag{10.1}$$

The coefficient is strictly positive at every original prime contracted by the prefix, and a nontrivial ordinary negative step has a nonzero comparison divisor. Discrepancies do not decrease. The transforms of Bott–Chern classes are linear. At a scaling wall the two scaled pullbacks agree.

Proof. These are the scaling and comparison statements of [42, Lemmas 3.6–3.7], with the ordinary-step conventions of Section 3.1 of that paper. □

10.2 The finite program

Proposition 10.2. *Let (X, B) be a smooth compact Kähler terminal pair with effective rational SNC boundary. Assume that $J = K_X + B$ is pseudoeffective and $N(J) \wedge B = 0$. Then there is a finite sequence of ordinary J-negative steps beginning at this X and ending at a compact Kähler terminal pair (M, B_M) with $K_M + B_M$ semiample. No boundary component is contracted. All working spaces are globally strongly $\mathbb{Q}$-factorial, and their underlying spaces are terminal.*

Proof. We apply Proposition 7.2 to the class of globally strongly $\mathbb{Q}$-factorial compact Kähler terminal pairs (T, B_T) with effective rational boundary, pseudoeffective $J_T = K_T + B_T$, and $N(J_T) \wedge B_T = 0$. Lemma 10.1 gives the scaling programs and comparisons; pseudoeffectivity persists by pushing forward a positive current representing the original adjoint on a common resolution.

We verify that the working pairs remain in this class. Strict comparison and exceptional translation place each contracted prime in the negative part of the current adjoint. Thus no boundary prime is contracted. On a common resolution, the same comparison shows that minimal multiplicities along surviving boundary primes remain zero. An exceptional valuation over the initial model has positive discrepancy, which cannot decrease. A prime of the initial model contracted later has boundary coefficient zero, and strict comparison raises its discrepancy from zero to a positive value. These are all exceptional valuations over a working model, so every working pair is terminal. Global strong $\mathbb{Q}$-factoriality makes its boundary $\mathbb{Q}$-Cartier; removing this effective boundary increases exceptional discrepancies, hence its underlying space is terminal as well.

Condition (i) of Proposition 7.2 is therefore satisfied, and Proposition A.2 supplies condition (ii). The criterion gives a finite ordinary program starting on X. Proposition A.1 makes its nef endpoint semiample. □

11 Preparation and deformation of logarithmic programs

We prepare the central pair in two steps: crepant stratum blowups make it terminal, and subtraction of the fixed boundary separates the remaining boundary from the divisorial negative part. Both steps preserve the prescribed central section space. The resulting finite logarithmic program then deforms over the disk.

Throughout this section the base is a disk and the fibers are connected. After shrinking, we split every boundary component and closed boundary stratum according to the connected components of its fibers. These are proper smooth families, so their sets of connected components are locally constant. Each resulting relative boundary component therefore restricts to one central prime divisor.

11.1 Simultaneous terminal preparation

The following terminal preparation refines the crepant stratum-blowup construction of [36, Section 4.2]. Keeping the coefficients in $m^{-1}\mathbb{Z}$ preserves the integral adjoint in the prescribed degree.

Lemma 11.1. *Let $f : \mathcal{X} \to \Delta$ be a smooth proper holomorphic family with compact Kähler fibers and relative SNC boundary $B = \sum b_i D_i$, where $0 \le b_i < 1$ and $mb_i \in \mathbb{Z}$ for a fixed positive integer m. After shrinking the disk there is a finite composition of blowups of smooth relative codimension-two boundary strata,*

$$\tau : \widetilde{\mathcal{X}} \longrightarrow \mathcal{X},$$

with a relative SNC boundary $\widetilde{B}$ such that

$$K_{\widetilde{\mathcal{X}}/\Delta} + \widetilde{B} = \tau^*(K_{\mathcal{X}/\Delta} + B), \qquad 0 \le \widetilde{B} < 1, \qquad m\widetilde{B} \text{ is integral.} \tag{11.1}$$

The new family is smooth with compact Kähler fibers. Its total pair and every fiber pair are terminal. Pullback identifies the degree-m adjoint sections, on the total spaces and on every fiber.

Proof. The blowups. Whenever two intersecting boundary components have coefficients b_i, b_j with $b_i + b_j \ge 1$, blow up a connected component of their intersection and give the new exceptional divisor coefficient $b_i + b_j - 1$. The center is smooth of relative codimension two, and the blowup formula gives (11.1) at this step. The new coefficient lies in $[0, 1)$ and is an integral multiple of $1/m$. The family stays smooth and the total boundary support, including any zero-coefficient exceptional components temporarily retained, stays relative SNC.

Termination. Consider the finite cone complex defined by the original relative SNC strata after shrinking around the compact central fiber. Each closed intersection has finitely many components, proper and smooth over the disk. In an original cone use its boundary generators as the lattice basis and put $d_i = 1 - b_i \ge 1/m$. The blowups above are star subdivisions along sums of two generators of a regular cone. Thus every inserted ray is primitive, has nonnegative integral coordinates $v = (v_i)$ in an original cone, and has log discrepancy

$$A(v) = \sum_i v_i d_i.$$

The coefficient assigned to that ray is $1 - A(v)$. It is inserted only when this coefficient is nonnegative, so

$$\sum_i v_i \le m.$$

There are finitely many such lattice vectors in each original cone. A subdivision inserts a ray not already present in the cone complex, and subsequent subdivisions retain every inserted ray. Keeping the finitely many original stratum labels distinguishes rays occurring over different connected components. The process therefore terminates.

Terminality and section comparison. At termination, any two intersecting positive boundary coefficients have sum less than one. Let v be an exceptional divisorial valuation over a smooth model in question, and take a general smooth point of its center, of codimension $c \geq 2$. Choose local coordinates such that $z_1, \dots, z_c$ vanish on the center and the boundary components containing it are $z_1 = 0, \dots, z_s = 0$. Write $v_i = v(z_i) \geq 1$ for $i \leq c$. Pulling back the coordinate volume form to a model carrying v shows

$$A_X(v) \geq \sum_{i=1}^{c} v_i :$$

in its exterior product at most one differential can lower the order by contributing the differential of a local equation of the exceptional prime. Consequently

$$a(v, X, B) \geq \sum_{i=1}^{s} (1 - b_i) v_i + \sum_{i=s+1}^{c} v_i - 1 > 0. \tag{11.2}$$

If $s \geq 2$, two of the first coefficients already sum to more than one. If $s = 1$, there is at least one further positive summand; if $s = 0$, the sum is at least two. This proves the strict inequality in every case. The same argument applies to the total smooth space and to every fiber; compare [30, Corollary 2.31].

Each fiberwise blowup is projective over a compact Kähler manifold and is again Kähler. Finally, multiplying the crepant identity by m gives an isomorphism of the integral adjoint line bundles. The equality $\tau_* \mathcal{O}_{\widetilde{\mathcal{X}}} = \mathcal{O}_{\mathcal{X}}$ and the projection formula give the section identification on the total spaces. The same argument applies to each fiberwise blowup. □

11.2 The fixed boundary in a prescribed degree

We next apply the degreewise fixed-part argument of [18, proof of Theorem 4.2]; compare [36, Section 4.3].

Lemma 11.2. *Let $(\mathcal{X}, B) \to \Delta$ be a prepared family as in Lemma 11.1. Suppose*

$$H^0\big(X_0, \mathcal{O}_{X_0}(m(K_{X_0} + B_0))\big) \neq 0.$$

There is a relative boundary $B_* \leq B$, with mB_* integral and with terminal total and fiber pairs, such that

$$J_* = K_{X_0} + B_{*,0} \text{ is pseudoeffective}, \qquad N(J_*) \wedge B_{*,0} = 0.$$

For the relative effective integral divisor $F = m(B - B_*)$, multiplication by its canonical section identifies the central section spaces:

$$H^0\big(X_0, m(K_{X_0} + B_0)\big) = s_{F_0}\, H^0\big(X_0, m(K_{X_0} + B_{*,0})\big). \tag{11.3}$$

Proof. Write e_i for the fixed multiplicity along $D_{i,0}$ of the complete central linear system, namely the least zero order there among its nonzero sections. Set

$$q_i = \min\{mb_i, e_i\}, \qquad F = \sum_i q_i D_i, \qquad B_* = B - F/m.$$

Both q_i and e_i are nonnegative integers. Every central section is divisible by s_{F_0}, and division produces a holomorphic section of the smaller line bundle. Conversely multiplication by s_{F_0} takes each section of that smaller line into the original complete system. This proves (11.3).

The new fixed multiplicity along $D_{i,0}$ is exactly $e_i - q_i$. If its remaining boundary coefficient $b_i - q_i/m$ is positive, then $q_i = e_i$. Some section of the smaller adjoint therefore has zero order along this prime. Its zero divisor divided by m represents a positive current in $\{J_*\}$ with zero generic Lelong number there. It follows that $\nu_{D_{i,0}}(J_*) = 0$. This proves the stated negative-part condition. The same nonzero section proves pseudoeffectivity of J_*. Lowering an effective boundary increases discrepancies, and all coefficients retain denominator dividing m. □

The relative divisor F records the central fixed multiplicities. Its section s_F restores the original boundary after extension for B_*.

11.3 Deformation of the prepared program

Proposition 11.3. *Apply Proposition 10.2 to the central pair obtained from Lemmas 11.1 and 11.2. After shrinking the disk, its finite ordinary program extends to proper relative bimeromorphic diagrams beginning at the prepared family $(\mathcal{X}, B_*)$. Their endpoint $g : (\mathcal{Y}, \mathcal{B}) \to \Delta$ has the following properties.*

(i) *$\mathcal{Y}$ is normal and g is proper and flat. Its central pair $(Y_0, \mathcal{B}_0)$ is the prescribed compact Kähler terminal endpoint. In particular $K_{Y_0} + \mathcal{B}_0$ is semiample.*

(ii) *The relative log adjoint $\mathcal{A} = K_{\mathcal{Y}/\Delta} + \mathcal{B}$ is $\mathbb{Q}$-Cartier. Both the total pair $(\mathcal{Y}, \mathcal{B})$ and the underlying total space are canonical near Y_0.*

(iii) *$\mathcal{B}$ is the horizontal strict transform of B_*, its coefficients lie in $[0, 1)$, and $m\mathcal{B}$ and $m\mathcal{B}_0$ are integral. Adjunction identifies $\mathcal{A}|_{Y_0}$ with $K_{Y_0} + \mathcal{B}_0$ as holomorphic $\mathbb{Q}$-line bundles.*

(iv) *A degree-m section on the endpoint total space transfers, under the meromorphic pluricanonical identification, to a section on the prepared family with boundary B_*. This transfer commutes with central restriction and with the analogous identification of central section spaces.*

Proof. Construction of the family. We apply [20, Theorem 7.7] with generalized nef datum zero. The prepared total space is smooth, the boundary is rational and horizontal, and the log adjoint is $\mathbb{Q}$-Cartier. After choosing a relatively compact smaller disk, the morphism is flat and proper with connected fibers. The central pair is canonical, has empty boundary floor, and satisfies $N(K_{X_0} + B_{*,0}) \wedge B_{*,0} = 0$ by Lemma 11.2. Thus the central hypotheses of that theorem hold.

Hacon–Xie extend the prescribed central ordinary steps. Their proof uses the construction of [16, Proposition 4.16], replacing the latter's projective-central-fiber model input by their Kähler model theorem. In that construction, contraction deformation gives the contraction diagrams; in the small case, the local relative log canonical models glue by uniqueness to give the flip. The construction identifies the central contraction, central flip, or central divisorial target with the prescribed one. It therefore gives proper relative diagrams with the prescribed normal central spaces. Since the program is finite, one shrinking of the disk suffices. The total spaces are normal by the construction. On each total space dominating the disk, the parameter is a nonzerodivisor. Torsion-freeness over the local discrete valuation ring of the base therefore gives flatness. This proves the family assertions in (i).

Boundary, singularities, and adjunction. A central divisorial contraction is an isomorphism at the generic point of every prime of its normal target; the two small contractions of a flip have

the same property. Lemma 2.5 extends each of these isomorphism loci to the total spaces. Applying it successively to the finite diagrams shows that their common isomorphism locus meets every endpoint central prime.

At these generic points, strict transform preserves the boundary coefficients and gives multiplicity-one central restrictions. Every endpoint central boundary prime is thus the transform of an original central boundary prime, and Proposition 10.2 preserves all such primes. The total boundary remains horizontal and its coefficients remain in $m^{-1}\mathbb{Z} \cap [0, 1)$.

The canonical-pair deformation in [16, Proposition 4.16], used by [20, Theorem 7.7], gives canonical total ordinary pairs and an $\mathbb{R}$-Cartier log adjoint near their central fibers. This applies to a normal Cartier central fiber with canonical restricted pair and boundary coefficients less than one; compare [7, Proposition 3.5] and the analytic Cartier-extension arguments in [29, Section 12.1]. In a local meromorphic canonical frame, the log adjoint is a rational Weil divisor. Since it is $\mathbb{R}$-Cartier, its local divisor class vanishes after tensoring with $\mathbb{R}$, hence also after tensoring with $\mathbb{Q}$. Thus the adjoint is locally $\mathbb{Q}$-Cartier. Compactness gives a common Cartier multiple near the central fiber.

Proposition 10.2 also makes each underlying central working space terminal. Deformation of canonical singularities therefore gives a $\mathbb{Q}$-Cartier relative canonical divisor and canonical underlying total space near the central fiber; see [28, Lemma 10 and Remark 10.3], based on [40, Theorem VI.5.2]. In particular the total space is klt.

At every generic central prime, the local isomorphism identifies ordinary adjunction and the multiplicity-one boundary restriction. Taking a common Cartier multiple extends that identification over the normal central fiber. This is the $\mathbb{Q}$-linear identification claimed in (iii). Lemma 2.2 consequently identifies $\mathcal{O}_{\mathcal{Y}}(m\mathcal{A})/t\mathcal{O}_{\mathcal{Y}}(m\mathcal{A})$ with $\mathcal{O}_{Y_0}(m(K_{Y_0} + \mathcal{B}_0))$. This proves (ii) and (iii).

Transfer of sections. Apply Lemma 2.4 to the canonical total pairs at the start and end of the diagrams. Their boundaries agree on common nonexceptional primes, and the prescribed degree is admissible on both sides. This gives the section transfer in (iv). The same lemma applies to the canonical central pairs. The two transfers commute with restriction on the common dense smooth open set, where they are ordinary pullbacks of relative pluriforms. The quotient identification above and reflexivity extend this commutativity to the whole central fiber. □

12 Proof of logarithmic invariance

Proof of Theorem B. Extension over a disk.

We first prove local extension over a disk. After shrinking, the connected components of a smooth proper family split, so we work with connected fibers. The zero-dimensional case is immediate.

Fix $m \geq 1$ and set

$$B^{(m)} = \frac{1}{m}\lfloor mB \rfloor. \tag{12.1}$$

This is a rational relative SNC boundary with coefficients in $[0, 1)$, and

$$\mathcal{L}_m = \mathcal{O}_{\mathcal{X}}\big(m(K_{\mathcal{X}/\Delta} + B^{(m)})\big). \tag{12.2}$$

For $m = 1$, the boundary $B^{(1)}$ is zero and Theorem D gives extension. For $m \geq 2$, it suffices to extend an arbitrary nonzero section $u_0 \in H^0(X_0, \mathcal{L}_m|_{X_0})$.

Apply Lemma 11.1 to obtain a crepant relative preparation

$$\tau : (\widetilde{\mathcal{X}}, \widetilde{B}) \longrightarrow (\mathcal{X}, B^{(m)}).$$

Lemma 11.2 then gives a boundary $\widetilde{B}_* \le \widetilde{B}$ and an effective relative integral divisor $F = m(\widetilde{B} - \widetilde{B}_*)$ such that

$$J_* = K_{\widetilde{X}_0} + \widetilde{B}_{*,0} \quad \text{is pseudoeffective,} \qquad N(J_*) \wedge \widetilde{B}_{*,0} = 0.$$

The boundary coefficients still have denominator dividing m. Proposition 10.2 supplies a finite central program with terminal semiample endpoint (Y_0, B_{Y_0}). Proposition 11.3 extends it to a proper flat family $g : (\mathcal{Y}, \mathcal{B}) \to \Delta$ and verifies the hypotheses of Theorem C, including adjunction and the integrality of $m\mathcal{B}$ and mB_{Y_0}.

The crepant preparation, fixed-boundary identity, and birational comparison give

$$\frac{\tau_0^* u_0}{s_{F_0}} \in H^0(\widetilde{X}_0, mJ_*) \simeq H^0(Y_0, m(K_{Y_0} + B_{Y_0})).$$

Theorem C extends the corresponding endpoint section in degree m. Transfer it back by Proposition 11.3(iv), obtaining

$$v \in H^0\big(\widetilde{\mathcal{X}}, m(K_{\widetilde{\mathcal{X}}/\Delta} + \widetilde{B}_*)\big).$$

Multiplication by s_F restores the boundary $\widetilde{B}$. The resulting section descends through τ by the integral crepant identity and the projection formula. Its central restriction is u_0. Thus every central section extends in its prescribed degree.

The line bundle $\mathcal{L}_m$ is flat over the disk, so Lemma 4.3 gives local constancy of $h^0(X_t, \mathcal{L}_m|_{X_t})$. Repeating the argument at every point proves constancy on a connected disk.

Smooth bases and relative consequences.

For a connected complex manifold S, restrict to coordinate disks in a coordinate polydisk. Each restricted family remains proper and smooth, with compact Kähler fibers and relative SNC boundary, and (12.2) restricts to its degree-m line bundle. The disk result makes the dimension constant along each such disk. Any two points of a smaller coordinate polydisk can be joined by finitely many such segments. The dimension is therefore locally constant, and connectedness makes it constant on S.

Grauert's theorem [15] now gives local freeness and, for every holomorphic map $a : T \to S$, the base-change isomorphism

$$a^*(f_*\mathcal{L}_m) \xrightarrow{\sim} (f_T)_*(p_{\mathcal{X}}^*\mathcal{L}_m), \qquad p_{\mathcal{X}} : \mathcal{X} \times_S T \to \mathcal{X}. \tag{12.3}$$

In particular the fiber of $f_*\mathcal{L}_m$ at s is $H^0(X_s, \mathcal{L}_m|_{X_s})$. A local frame therefore extends each fiber section over a neighborhood of s. If S is Stein, Cartan's theorem A makes global sections of $f_*\mathcal{L}_m$ surject onto each fiber. Since $H^0(S, f_*\mathcal{L}_m) = H^0(\mathcal{X}, \mathcal{L}_m)$, the extension can then be taken over all of $\mathcal{X}$. □

13 Twistor families and the Kähler hypotheses

The twistor construction gives smooth families whose fibers are compact Kähler and whose total spaces are non-Kähler over every open subset of the base. The period description makes this local property explicit.

Proposition 13.1. *Let $q : Z \to \mathbf{P}^1$ be the twistor family of a hyperkähler K3 surface. For every nonempty connected open subset $U \subset \mathbf{P}^1$, the restriction $Z_U \to U$ is a proper holomorphic submersion with compact Kähler fibers, whereas Z_U is not Kähler.*

Proof. Let M be the underlying smooth four-manifold, with hyperkähler metric g and quaternionic complex structures I, J, K. Put

$$\omega_I = g(I\,\cdot,\cdot), \quad \omega_J = g(J\,\cdot,\cdot), \quad \omega_K = g(K\,\cdot,\cdot).$$

The twistor construction gives the differentiable product $M \times \mathbf{P}^1$ an integrable complex structure for which the second projection is a holomorphic submersion. The fiber indexed by $L = aI + bJ + cK$, $a^2 + b^2 + c^2 = 1$, has complex structure L and Kähler form

$$\omega_L = a\omega_I + b\omega_J + c\omega_K.$$

We use the standard construction and its period description from [23, Chapter 7, Section 3.2]; see also [24, Section 1.3]. Compactness of M gives properness, including after restriction to U.

Under the differentiable identification of the fiber cohomologies, let

$$W = \operatorname{span}_{\mathbb{R}}\{[\omega_I], [\omega_J], [\omega_K]\} \subset H^2(M, \mathbb{R}).$$

The intersection form is positive definite on W. Indeed quaternionic linear algebra gives

$$\omega_I \wedge \omega_J = \omega_J \wedge \omega_K = \omega_K \wedge \omega_I = 0, \qquad \omega_I^2 = \omega_J^2 = \omega_K^2 = 2\,\mathrm{vol}_g\,.$$

For the complex structure L, the real $(2,0) + (0,2)$ plane is

$$P_L = W \cap [\omega_L]^{\perp}. \tag{13.1}$$

For example, complete L to an oriented orthonormal quaternionic triple L, L', L''; the holomorphic symplectic form is $\omega_{L'} + \sqrt{-1}\omega_{L''}$, proving (13.1).

Suppose that Ω were a Kähler form on Z_U. Since it is closed and the fiber inclusions in $M \times U$ are homotopic along paths, their restrictions have a fixed de Rham class $\alpha \in H^2(M, \mathbb{R})$. Each restriction is of type $(1,1)$ for its fiber. Hodge orthogonality and (13.1) therefore imply that the W-projection of α lies in $\mathbb{R}[\omega_L]$ for every $L \in U$. The open set U contains two distinct nonantipodal points of the sphere, whose two lines have zero intersection. Thus this projection is zero, and

$$\int_M \alpha \wedge [\omega_L] = 0 \qquad (L \in U).$$

On the other hand, $\Omega|_{(M,L)}$ and ω_L are Kähler forms on a compact complex surface. Their wedge is a strictly positive volume form, so this integral is positive. The contradiction proves the proposition. □

Example 13.2 (A nonzero relative boundary). Let U be a disk in the twistor sphere, let C be a smooth projective curve of genus $g \geq 2$, and choose $r \geq 1$ distinct points $p_1, \ldots, p_r$ and rational coefficients $0 < b_i < 1$. The family

$$\mathcal{X} = Z_U \times C \longrightarrow U, \qquad D_i = Z_U \times \{p_i\}, \qquad B = \sum_i b_i D_i$$

has compact Kähler fibers and a nonzero relative SNC boundary with disjoint components. Its total space is not Kähler over any smaller nonempty disk: restriction of a hypothetical Kähler form to $Z_U \times \{p\}$ would contradict Proposition 13.1.

These examples have nonzero log pluricanonical spaces in every degree. The canonical line of every K3 fiber is trivial, and its pluricanonical space is one-dimensional. The product formula for sections therefore gives

$$h^0\big(X_t, mK_{X_t} + \lfloor mB_t \rfloor\big) = h^0\left(C, mK_C + \sum_i \lfloor mb_i \rfloor p_i\right).$$

For $m = 1$ the right side is g. For $m \geq 2$, Serre duality gives zero first cohomology, since $(1-m)K_C - \sum_i \lfloor mb_i \rfloor p_i$ has negative degree. Riemann–Roch gives the explicit value

$$(2m-1)(g-1) + \sum_i \lfloor mb_i \rfloor.$$

The formula computes the logarithmic plurigenera of this family in every degree and exhibits their constancy on U.

A Abundance and nef comparison

The termination arguments use two consequences of [42]. Its abundance theorem [42, Theorem 2.13] applies by logarithmic subadditivity [41, Corollary 6.2]. We state the resulting decomposition and nef comparison in the form used here; all decompositions are $\mathbb{Q}$-linear equivalences of holomorphic $\mathbb{Q}$-line bundles.

Proposition A.1 (Good divisorial decomposition)**.** *Let (X, B) be a normal compact Kähler log canonical pair, with B an effective rational divisor and $J = K_X + B$ a pseudoeffective $\mathbb{Q}$-Cartier adjoint. There is a smooth compact Kähler modification $\mu : W \to X$ such that*

$$\mu^* J \sim_{\mathbb{Q}} P + N(\mu^* J), \qquad P \text{ semiample},$$

and $N(\mu^* J)$ is rational. On every higher smooth model $a : W' \to W$, the negative part is $a^* N(\mu^* J)$.

Proof. Apply [42, Theorem 2.13], with the higher-pullback and normal-pair formulations of [42, Lemma 2.4 and Proposition 2.7]. □

In particular, a nef adjoint in Proposition A.1 is semiample: its negative part vanishes after pullback, and generation of a Cartier multiple descends by normality and the projection formula.

Proposition A.2 (Nef comparison near zero)**.** *Let (T, B_T) be a normal compact Kähler klt pair with effective rational boundary, and put $J_T = K_T + B_T$. Suppose that J_T is $\mathbb{Q}$-Cartier and pseudoeffective, and that $N(J_T) = 0$ on the prime divisors of T. For every Kähler class A there are a normal compact Kähler space V, a small bimeromorphic map $T \dashrightarrow V$, transformed classes $\{J_V\}, A_V$, and $\delta > 0$ such that*

$$\{J_V\} + sA_V \quad \text{is nef for } 0 \leq s \leq \delta.$$

On a common smooth resolution, the difference of the original and transformed pullbacks is represented by a real divisor exceptional over both models, depending affinely on s.

Proof. This is the empty-floor case of [42, Lemmas 3.11–3.12]. Those lemmas apply to normal dlt pairs. Here the pair is klt, so their auxiliary divisor supported on $\lfloor B_T \rfloor$ is zero. Their dimensional hypothesis is satisfied with $n = \dim T + 1$, since [42, Theorem 2.13] supplies good decomposition in every dimension. □

The condition $N(J_T) = 0$ concerns the prime divisors of T, which is exactly the vanishing obtained in Proposition 7.2 before restarting the scaling program.

AI Disclosure

All mathematical derivations and proofs were produced by AI. The human author formulated the problem, provided relevant literature, and guided the research direction. AI also assisted in the selection of some of the problems.

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